

The Performance of Venture-Backed IPOs in Germany: Exit Strategies, Lock-up Periods, and Bank Ownership

Wolfgang Bessler
Andreas Kurth

Center for Finance and Banking
Justus-Liebig-University Giessen

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Address: Prof. Dr. Wolfgang Bessler, Center for Finance and Banking, Justus-Liebig-University Giessen, Licher Strasse 74, D-35394 Giessen. Tel: (49) 641 99 22 461, Fax: (49) 641 99 22 469, email: Wolfgang.Bessler@wirtschaft.uni-giessen.de

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ABSTRACT

The agency problems for initial public offerings are well documented in the literature. The objective of this research is to investigate the potential conflicts of interest for the "Neuer Markt" in Germany. Of special interest are venture-backed IPOs and those in which banks acted as venture capitalist, underwriter, and provided analyst recommendations. We observe high initial returns and out-performance over the first 6 months of trading which decreases significantly over the subsequent 18 months. The individual performance depends on the VC's underwriter and bank affiliation, exit behavior, and lock-up commitment. Venture capitalists and especially banks timed their exit well. This indicates some serious agency problems in the German IPO market.

Keywords: Initial Public Offerings, Venture Capital, Conflicts of Interest, Bank Equity Ownership

JEL Classification: G 14; G15; G24; G32

1. Introduction

The significance of venture capital for predicting the risk and return of initial public offerings (IPOs) has been investigated extensively in numerous theoretical and empirical studies (Gompers and Lerner, 1999a). The basic argument is that venture capitalists possess certain screening and monitoring abilities that positively impact initial and long-run returns and consequently the market value of the firm. This notion can be empirically tested by comparing the financial performance of ventured-backed and non venture-backed IPOs. Several empirical studies for venture-backed IPOs find first of all a smaller underpricing and second a smaller long-run underperformance for venture-backed IPOs (Megginson and Weiss, 1991; Brav and Gompers, 1997). This supports the idea that venture capitalists are in a position to reduce the adverse selection problems for IPOs by signaling the quality of the firm. Moreover, venture capitalists perform important corporate governance functions in that they force management to act in the interest of shareholders. Nevertheless, they are first of all interested in their own profitable exit which may lead to specific exit strategies and conflicts of interest.

The objective of our analysis is to provide empirical evidence on the impact of venture capital financing and especially on bank involvement on underpricing and long-run performance of IPOs in Germany's "Neuer Markt". Because venture capital firms and especially banks follow their own agenda in maximizing their market value, one would expect that a number of severe agency conflicts arise from these relationships. This may be especially true for the German universal banking system where financial institutions traditionally hold equity stakes in firms, provide venture capital financing and loan services before and after the IPO but also offer full or partial underwriting services and securities trading. In addition, the banks in Germany are involved in mutual fund management as well as in investment research and analyst recommendations. Consequently, the financial market participants are faced with a peculiar situation when a bank that is the lead underwriter or is part of the underwriter syndicate also holds a venture capital stake in the firm before the IPO. For the underwriter the due diligence as well as the bookbuilding process offers new informational insights that can be either signaled to the market or exploited. The information advantage that the bank acquires as a venture capitalist may also

lead to either a better signaling of the quality of the firm (certification hypothesis) or to severe conflicts of interest between the underwriter and potential investors (conflict-of-interest hypothesis) when the bank exploits its private information. One very recent example for these ongoing agency problems in the German market is Deutsche Bank which acted as the lead underwriter for the Postbank IPO in 2004. After completion of the due diligence process and during the pricing stage there were rumors that Deutsche Bank was interested in buying the bank instead of bringing it to the market. In the end Deutsche Bank judged Postbank as too expensive and abstained from an offer which did not help the stock price during the going public process (Börsenzeitung, June 21, 2004). Another example is Baader Wertpapierhandelsbank which recommended buying shares of the Softing AG several times even before the date of the IPO. This bank was a major shareholder (via a subsidiary) of this company and the lead underwriter (IPO prospectus). These examples reveal that several relationships and interactions will have to be investigated in order to gain a complete understanding of the conflicts of interest in the German universal banking system with respect to IPOs. One reason why the German market for new issues compared to other markets did not rebound yet from the crash in 2001 could be related exactly to the recent experience with these agency conflicts that keeps the investor away from such equity offerings.

These various issues will be empirically tested by analyzing the performance of venture-backed firms that went public at the “Neuer Markt” during the period from 1998 to 2001. We further investigate bank controlled venture capital firms separately because the agency conflicts as mentioned above may be most pronounced. Of special interest are those venture-backed IPOs for which the VC also acted as an underwriter in the going public process. The results for the “Neuer Markt” could differ from other markets for growth companies and offer new insights due to several special listing requirements such as a mandatory lock-up period of six months and the required issuance of new primary shares (minimum 50% of issue size) at the time of the IPO. The empirical findings for IPOs in Germany clearly indicate that there are substantial conflicts of interest when banks provide venture capital financing, act as underwriter, and provide investment advice. First of all these IPOs are significantly higher underpriced. Furthermore, we

find for bank venture-backed IPOs evidence of successful exit timing. However, if banks sell a major part of their stake already at the time of the IPO, abnormal returns drop immediately below average. Thus, the market seems to be aware of agency problems. In addition, banks are not able to mitigate these problems by voluntarily extending their lock-up period. Nevertheless, banks seem to be in a position to provide profitable exit opportunities by issuing positive analyst recommendations. Our empirical findings are in line with the results of Ber et al. (2001) for Israel who provide evidence of agency problems when a financial institution acts as commercial bank, underwriter, and manages mutual funds. Thus, the significance of agency problems seems to depend on the set up of the financial system and the German universal banking system may be especially exposed to such conflicts of interest. Consequently, changes are necessary to create a successful equity market for start-up firms in Germany.

The rest of the paper is organized as follows. In the next section the literature with respect to agency problems and performance of IPOs is reviewed. Section 3 contains a description of the data and methodologies. The results for the underpricing and long-run performance are presented in section 4. There we distinguish between venture-backed and bank-financed IPOs, between different lock-up periods, and analyze the impact of analyst recommendations. Finally we validate our empirical findings with a cross sectional regression analysis. The last section concludes the paper.

2. Literature review

There exists a substantial amount of literature investigating the underpricing and long-run underperformance of initial public offerings for the U.S. and other countries (Ritter, 2003). For a more detailed understanding of initial returns and long-run performance it is important, however, to analyze the impact of specific factors such as venture capital (2.1), lock-up periods (2.2), and underwriting by commercial banks (2.3) as well as bank underwriting in conjunction with venture capital investments (2.4) and analyst recommendations (2.5). In this section the literature for these aspects is reviewed and the interesting questions and unsolved issues for an empirical analysis of the “Neuer Markt” in Germany are presented.

2.1. *Venture capital*

Common knowledge suggests that venture capital firms should contribute significantly to the success of a company. This is usually attributed to the VC abilities of screening, monitoring, and consulting. Thus, a VC involvement should better signal the quality of an IPO leading to a smaller underpricing and a superior long-run performance. Nevertheless, the empirical findings for these questions are mixed so far. For venture-backed IPOs in the U.S. Megginson and Weiss (1991) find on average a lower underpricing whereas Barry et al. (1990) observes this only for experienced VCs. Francis and Hasan (2001) do not find a lower underpricing and Lee and Wahal (2004) even document higher initial returns for venture-backed IPOs due to the grandstanding argument provided by Gompers (1996). With respect to the long-run performance of venture capital firms Brav and Gompers (1997) report superior returns and Lerner (1994) finds a special ability of VC to time their exit well. This latter result is confirmed by Gompers and Lerner (1998) who observe that venture-backed IPOs significantly outperform before the exit and significantly underperform after the exit of the venture capitalist. These empirical results may suggest some exceptional insights or abilities by venture capitalists with respect to valuation and exit behavior. In contrast, such a performance may be the result of private information or specific price influencing measures, indicating that certain conflicts of interest may arise when venture capital firms are involved.

With respect to the “Neuer Markt” in Germany Kiss and Stehle (2002) find initial returns of 50% compared to 17.5% for the “Nouveau Marché” in France, 23.5% for the “Nuovo Mercato” in Italy (Giudici and Roosenboom, 2002), and 39.0% for the “SWX New Market” in Switzerland (Drobetz et al., 2003). Franzke (2001) reports for venture-backed IPOs in Germany an even higher underpricing of 64.6% in comparison to 61.2% for non venture-backed IPOs. It is interesting to note that the granting of an over-allotment option (greenshoe) does not have any significant impact on price stabilization activities or aftermarket performance (Franzke and Schlag, 2002). The long-run performance for IPOs in Germany before the opening of the “Neuer Markt” (1970-1995) was investigated by Bessler and Thies (2004). They find the usual long-run underperformance (36 months) with the exception that IPOs with subsequent season

equity offerings (SSEO) outperformed the market significantly. For the “Neuer Markt” Kraus and Burghof (2003) report that venture-backed compared to non venture-backed IPOs have quite a similar performance after 6 months but inferior abnormal returns after 18 months. Thus, the length of the lock-up period seems to be the determining factor.

2.2. Lock-up period

A significant impact of IPO lock-up periods is reported in studies for the United States (Field and Hanka, 2001), the United Kingdom (Esenlaub et al., 2001), and Germany (Nowak and Gropp, 2002). All studies report a negative performance around the end of the lock-up period at which time the venture capital firms seem to have exited. In addition, Brav and Gompers (2003) find extended lock-up periods for those IPOs that may have some severe agency problems. Hence, lock-up periods may also be employed as a positive signaling device. Goergen et al. (2004) report similar evidence for French and German IPOs by analyzing the lock-up contracts. However, they do not investigate the impact on the long-run performance as is done in our study. Thus, it is interesting to investigate whether longer lock-up periods indeed signal IPO quality and lead to significantly higher long-run returns. Of special interest are possible agency problems of venture financing in general and in particular when the venture capitalist is affiliated with a bank. In this case the special certification role of the venture capitalist documented above may be overshadowed by other factors or agency problems.

2.3. Underwriter

The reputation and behavior of the underwriter play a crucial role in the going public process and may significantly impact initial and long-run returns. Possible conflicts of interest have always been a concern and the solution for the U.S. was a separation of commercial and investment banking activities in different legal entities (Glass-Steagall Act of 1933) as well as strong securities regulation (SEC). This strict separation was slowly faced-out during the 1980s and finally abandoned in 2000. For the period before 1933 most empirical studies of security issuance for the U.S. could not confirm the suggested agency problems between the loan and underwriting businesses (Ang and Richardson, 1994; Kroszner and Rajan, 1994; Puri, 1996) and

the economic justification for the Glass-Steagall Act. Studies for the more recent periods during which most U.S. commercial banks participated again in underwriting find a lower underpricing for commercial bank underwritten IPOs (Hebb, 2002), suggesting that banks play a positive certification role due to their private information. Fields et al. (2003) also report lower underpricing and a superior long-run performance for commercial bank underwritten IPOs. Thus, commercial banks as underwriters seem to fairly price IPOs in the United States (Fields and Fraser, 2004). Schenone (2004) also suggests that underwriters were able to certify the quality of IPOs if they had provided loans several years before the IPO. The lower underpricing for commercial bank underwritten IPOs is also confirmed in studies by Ursel and Ljucovic (1998) for Canada. All these studies suggest a lower underpricing for IPOs underwritten by banks that provide both lending and underwriter services, indicating a credible signaling mechanism. Thus, banks possess private information from the loan business and therefore are able to reduce the information asymmetry between the firm and the investor. However, when bank information and especially underwriter information is paired with venture capital investments this may be the basis for substantial agency problems.

2.4. *Underwriter and venture capital*

As usual, the use of private information may be the source of special conflicts of interest. Gompers and Lerner (1999b) investigate for venture-backed IPOs in which the venture capitalist and the underwriter belong to the same financial institution whether possible agency problems are correctly anticipated by the investor at the time of the IPO resulting in a higher underpricing (rational-discounting hypothesis). In contrast, lower initial returns are expected if the underwriter exploits his information advantage (naive-investor hypothesis). After the IPO both groups should perform similar. The empirical findings for the U.S. confirm the existence of a higher underpricing but find no significant underperformance afterwards. It appears that the potential agency problems are fully anticipated by the U.S. investor. In contrast, Li and Masulis (2003) find a lower underpricing and could not confirm the existence of severe conflicts of interest between venture financing and underwriting IPOs. They conclude that the lead underwriter reliably signals the quality of the IPO, thus reducing agency problems. For the UK

Espenlaub et al. (1999) report similar findings whereas Hamao et al. (2000) document opposite results for Japanese IPOs underwritten by affiliated VC firms. These IPOs are more underpriced and issued with higher PE-ratios, indicating overvaluation at the time of the IPO. Thus, it appears that the empirical findings seem to depend to some extent on the banking system and the efficient regulation of securities markets.

2.5. Underwriter and analyst recommendations

Moreover, the underwriter usually has various means to influence the stock prices of IPOs in the primary and secondary market, thus impacting the short- and long-run performance of these firms. Underwriter or banks with an equity stake, for example, may convince the analyst to issue too favorable investment reports and buy recommendations. Lin and McNichols (1998) as well as Michaely and Womack (1999) find support for this idea in the U.S. in that the recommendations by analysts belonging to the underwriter are positively biased. Moreover, price support by the bank or its mutual fund is not easily observable but likely (see Ber et al. 2001 for Israel). The interest and timing of an extraordinary positive but only temporary market performance is obviously related to the invested capital of the venture capital firm and the equity stake of the bank as well as the length of the lock-up period. All this influences the exit behavior of institutional investors. These various issues discussed in this section with respect to venture capital firms and banks will be empirically tested for the German capital market as outlined in the next section.

2.6. Research agenda

This review of the literature provides important insights into the impact of venture capital financing on initial returns and long-run performance of IPOs as well as on the potential conflicts of interest when the same financial institution provides venture capital financing, underwriting services, and issues analyst recommendations. This is usually the case in the German universal banking system. The research agenda for the empirical analysis is presented in Figure 1 and outlined below. First of all the impact on venture capital financing is analyzed in section 4.1. The certification hypothesis predicts that venture-backed IPOs should be less underpriced

and have a superior long-run performance. However, the opposite could be true due to specific IPO characteristics such as lower and more volatile earnings, less tangible assets and higher expected growth rates, as well as the grandstanding argument and the expected optimal exit timing by the venture capitalist.

[Please insert Figure 1 about here]

In section 4.2 we investigate the special conflicts of interests that can arise when the VC is also the underwriter of the IPO. If these conflicts of interests are correctly anticipated by the market a higher underpricing should be observed but no difference in long-run performance. An opposite result could be expected if the underwriter uses his private information to act in the interest of the investor and signals the true quality of the IPO (4.2.1). If the venture capital, however, is provided by a commercial bank instead of an independent VC, there are additional factors that could influence the IPO performance. On top of the underwriter argument banks could try to influence stock prices through analyst recommendations, to time their exit around the highest stock price, allocate the lemons or the cherries to specific mutual funds, and finally make sure that outstanding loans are repaid first. In this case building reputation would not be a priority of the bank. This could result in a higher underpricing and a superior long-run performance (4.2.2).

Lock-up periods (4.3) are a costly commitment of the shareholder at the time of the IPO because the first exit opportunity is postponed to a later date. The legally required period at the “Neuer Markt” was six months but could be extended to a longer period. Thus, if the VC keeps the shares voluntarily for an extended lock-up period this could be interpreted as a positive signal about the quality of the IPO (4.3.1). In contrast, an extended lock-up period could also be misused by banks for the reasons given in the previous paragraph (4.3.2).

The exit behavior of the venture capitalist (4.4) could also be interpreted as a signal about the quality and future performance of the IPO. Due to substantial positive returns during the first day of trading (initial returns), VCs usually would not sell a major part of their shares in the primary market although this was legally possible. In the secondary market, however, they are restricted for the lock-up period for at least six months. Thus, major VC sales at the time of

the IPO could signal an expected weak IPO performance, thus resulting in a higher underpricing and an inferior long-run performance. Because of the potential conflicts the empirical findings could differ for independent VCs (4.4.1) and banks (4.4.2). Finally, buy recommendations by affiliated analysts could positively bias the stock price performance for some time to allow for a favorable exit, e.g., at the end of the lock-up period. Thus, the performance especially of bank-financed IPOs could be positively influenced by analysts (4.5).

3. Data and methodology

3.1. Data

The German “Neuer Markt” segment for growth companies was opened in 1997 and was finally closed in 2003 due to some legal and performance problems. The sample includes all IPOs for the “Neuer Markt” for the period from 1998 to 2001. The 12 IPOs in 1997 were excluded because they are not growth companies and are more similar to the firms that went public before 1997. There was one IPO in 2002 and none in 2003. 7 IPOs had to be excluded from the sample due to the unavailability of data (bankruptcy and take over). Moreover, two possible stock manipulation cases were excluded as well (COMRoad AG and Infomatec AG). Thus, the final sample consists of 307 companies for which all necessary data as well as stock returns for the first 24 months of trading is available. Share prices were gathered from Reuters. Although the “Neuer Markt” was closed as a market segment, it offers a great opportunity to analyze the agency problems that are related to the going public process in the German universal banking system which may have been one reason for the problems and the final closing.

The focus of this research is on the investment and exit strategies of venture capitalists and in particular of IPOs in which banks were involved as venture capitalist. Thus, we employ a broad as well as a narrow definition of VC financing. In the broad definition, we define venture capital firms as institutional shareholders that focus on start-up financing, including bank subsidiaries. To be included, they either have to be a member of a VC organization or they have to be classified as a professional start-up financier on its homepage. The ownership structure and the lock-up periods are gathered from each IPO’s issuance prospectus. We aggregate the VC

shareholdings and classify those IPOs as venture-backed when the pre IPO cumulative VC ownership exceeds 4%. This cut-off point was chosen because there seems to be natural break point in data and a higher threshold would have reduced the sample size. Applying this definition 101 IPOs are venture-backed. In order to analyze the potential conflicts of interests in venture financing in more detail we divide the 101 venture-backed IPOs into several sub-samples. The first criterion we use is whether a VC is part of the IPO underwriter syndicate. We observe 35 underwriter-affiliated and 66 not underwriter-affiliated venture-backed IPOs. In 22 cases the VC acts even as the lead underwriter.

In the narrow definition of venture capital we consider only independent venture capitalists, i.e., bank subsidiaries are excluded. Again, we apply a cut-off rate of 4% pre IPO stock ownership. Overall there are 72 IPOs in which an independent (non-bank) VC owns more than 4% of the shares before the IPO (panel B). In addition, we examine those IPOs which were financed by a subsidiary of a (commercial) bank. Again, a cut-off rate of 4% pre IPO ownership is applied. There were 44 venture-backed IPOs in which a bank was identified as a major shareholder (bank-financed IPOs, panel C). 15 IPOs were financed both by non-bank and bank VCs.

[Please insert Table 1 about here]

Of special interest is the VC's exit behavior, i.e., its equity ownership before and after the going public date. The data is presented in Table 1. In panel A banks are included. Before the IPO venture capital firms own on average 28.34% of the shares which reduces to 15.96% after the IPO. On average they sell almost 25% of their share holdings at the IPO date. The remaining shares are locked-up for at least six months. In 27 cases (of 101) the VC agreed to a longer lock-up period. From panels B and C it is evident that banks own fewer shares before going public (16.83%) and sell more shares at the IPO date (27.88%). In contrast, independent VCs hold 29.21% of IPOs shareholdings and sell 24.22% of their stock in the primary market. Thus, there seems to be some difference in exit behavior between banks and independent VCs.

3.2. Methodology

We calculate IPO initial returns (underpricing) by relating the first price in the second-

dary market to the issuing price. The long-run performance is measured by holding period return (BHAR) which are calculated for a single stock i for time period T as follows:

$$(1) \quad BHR_{i,T} = \left[\prod_{t=1}^T (1 + R_{i,t}) \right] - 1$$

where $R_{i,t}$ is the return of stock i at time t , and T is the time period for which the BHR is determined. For an equally-weighted portfolio of stocks the returns is calculated as:

$$(2) \quad dBHR_{p,T} = \frac{1}{N} \sum_{i=1}^N BHR_{i,T}$$

where $dBHR_{p,T}$ is the average BHR of the portfolio, N is the number of stocks in the portfolio, and T is the time period for which the BHR is calculated. To calculate “Buy-and-Hold-Abnormal>Returns” (“BHAR”) the return of the benchmark is subtracted from the IPO return.

$$(3) \quad BHAR = \frac{1}{N} \sum_{i=1}^N \left[\left(\prod_{t=1}^T (1 + R_{i,t}) \right) - \left(\prod_{t=1}^T (1 + R_{M,t}) \right) \right]$$

The advantage of this method is that the terminal values of the two strategies, i.e., investing in an IPO or investing in the benchmark, are directly comparable. Thus, BHAR compare real investment strategies over a defined period. The “NEMAX Allshare” index is used as a benchmark. This market weighted performance index includes all companies that are listed at the “Neuer Markt”.

We test for the statistical significance of our empirical findings with various methodologies. Because abnormal returns are highly skewed both parametric and non-parametric tests are employed. To analyze whether the mean BHAR is significantly different from zero we employ a skewness-adjusted t-test as documented by Lyon et al. (1999). In addition we check whether the number of firms with positive excess returns is larger than the number of firms with negative excess returns by employing a sign-test. Furthermore, we compare the BHAR of two samples. To test whether the abnormal returns of the two samples differ significantly we apply again a parametric t-test for the differences between means and a non parametric Mann-Whitney U-test.

4. Empirical results

First of all, we examine the underpricing and long-run performance of venture-backed and non venture-backed IPOs for the entire sample (4.1). In the following section (4.2.), the venture-backed IPOs are divided into various sub-samples, i.e., whether the VC is affiliated with a bank or is a member of the underwriter syndicate. Furthermore we investigate the exit behavior of independent VCs and banks by the length of the committed lock-up period (4.3) and the number of shares sold at the time of the IPO (4.4). For the group of bank-financed IPOs it is analyzed (4.5) whether recommendations by affiliated analysts have any impact on the IPO performance. In the last section of this chapter (4.6), we summarize our major empirical findings so far. In chapter 5, these relationships and agency problems are tested with various regressions in which market values and market-to-book ratios are included as control variables.

4.1. *Venture-backed versus non venture-backed IPOs*

In the very general part of the empirical analysis we investigate the underpricing and performance of all 101 venture-backed IPOs in our sample. These results are presented in Table 2 and Figure 2. Venture-backed IPOs are slightly more underpriced (55.0%) than non venture-backed IPOs (47.1%) and after the first 6 months of trading the performance (BHAR) of venture-backed IPOs (36.0%) exceeds the performance of the non venture-backed IPOs (24.5%). Interestingly, after 24 months the difference in performance narrows and nearly vanishes for the two groups to 6.5% and 3.0%, respectively. As a first result, initial returns and BHAR after 6 months of trading are significantly positive for both samples. However, tests for the difference in mean returns and the U-tests are not statistically significant. These empirical findings suggest that there is no apparent positive impact of venture capital financing on the underpricing and long-run performance of IPOs in Germany. Nevertheless, it is interesting to note that a clear return pattern emerges over time in that the difference of BHAR between the two groups increases for the first 6 months, then narrows for the subsequent 6 months, and then stays constant for the following 12 months. This pattern indicates the importance of the lock-up period which is examined in a later section.

[Please insert Figure 2 and Table 2 about here]

4.2. *Venture-backed IPOs, underwriter affiliation and bank ownership*

For a more detailed analysis the sample of 101 venture-backed IPOs is now divided into two sub-groups. First, we consider whether the VC is a member of the underwriter syndicate (4.2.1) and second, whether the VC is affiliated with a bank (i.e., bank-financed) or independent (i.e., venture-backed in a narrow sense; 4.2.2).

4.2.1. *Venture-backed IPOs and underwriter business*

There are 35 cases in the “Neuer Markt” in which the venture capitalist is part of the underwriter syndicate. It may be fair to assume that these venture capital providing banks have some influence on the pricing and allocation of shares at the time of the IPO. The initial returns of these IPOs are quite substantial in that we find significant returns of 84.1% for underwriter-affiliated venture-backed IPOs and 39.6% for the other venture-backed IPOs (Table 3). This difference is statistically significant at the 5% level. A comparison of the long-run performance (6 and 24 months), however, does not reveal any significant difference for these two groups (Figure 3 and Table 3).

[Please insert Figure 3 and Table 3 about here]

When these empirical findings for the “Neuer Markt” in Germany are compared to studies for the U.S., they are quite similar to the results of Gompers and Lerner (1999b) but are in contrast to the results of Li and Masulis (2003). It could be argued that the expected conflict of interest in the German banking system requires a higher underpricing which is enforced during the bookbuilding process. Thus, it could also be argued that such a higher underpricing is to the disadvantage of the affiliated IPO - which leaves money on the table - but to the obvious advantage of the investors who gets these underpriced shares allocated (Ljungqvist and Wilhelm, 2003). This profit allocation is even more of a concern when these shares are allocated to an affiliated mutual fund. Baums and Theissen (1999) and Löffler (2003) find some evidence of such a bank behavior for the German market.

The major question that arises from this analysis is whether German universal banks act

as underwriters in the specific interest of their corporate customers by helping them to increase the offer price at the time of the IPO as well for a certain time period afterwards in the secondary market with price support measures and analyst recommendations or whether they act in the interest of specific investors or investor groups by allocating shares with great upside potential at relatively low prices to a preferred investor clientele. Usually one would expect that the banks' duty is to act as an efficient intermediary between surplus and deficit spending units, i.e., investors and companies, respectively, by setting fair and market clearing prices (Puri, 1999). Because banks as private equity investor may have certain interests on their own - which may lead to various agency problems - it is useful to investigate the performance of bank-financed IPOs in more detail.

4.2.2. Bank-relationship and IPOs

There are 44 firms of 101 venture-backed firms at the "Neuer Markt" in which banks had a private equity stake of more than 4% before the time of the IPO. The empirical results are summarized in Table 4 and Figure 4. We find significant initial returns of 73.6% for the bank-financed IPOs and of 40.7% for the other venture-backed IPOs. Again this difference in mean returns is significant at the 5%-level indicating some possible conflict of interest in bank equity financing. At a first glance there does not appear to be any obvious difference in long-run performance for both sub-samples. For the first 6 months the BHAR is 26.4% for the bank-financed IPOs and 43.5% for the non-bank financed IPOs. After 24 months the BHAR are – 11.1% and 20.1%, respectively. The difference in mean returns appears to be substantial but is not significant.

[Please insert Figure 4 and Table 4 about here]

4.3. Impact of the lock-up period

The official regulation of the German Stock Exchange requires a non-negotiable lock-up period of at least six months for the "Neuer Markt". Although lock-up periods are perceived as quite important in signaling the quality of an IPOs it is in most countries voluntary and rests on a contract between the underwriter and the firm. Given the mandatory lock-up period of at

least six months in Germany the time window around the expiration date is of special interest. In addition, this is often the time period during which the required lock-up period for the venture capitalist ends. From this point on they are in a position to exit the venture by selling their equity stakes in the secondary market. In some cases, the venture capitalists agreed voluntary or involuntary on longer lock-up periods. Such a behavior when VC initiated or voluntary could signal a commitment and positive outlook for the IPO. It could also signal quality problems if the extension was demanded by the underwriter or committed by a bank. Therefore, it is useful to separate the sample into two groups in order to investigate the performance of IPOs depending on the length of the lock-up period. We apply this procedure for both non-bank (4.3.1) and bank venture-backed IPOs (4.3.2).

4.3.1. *Venture-backed IPOs and lock-up commitment*

The 57 venture-backed IPOs without any bank ownership are divided into two subsamples. In 12 cases, the VC committed to an extended lock-up period of more than six months. In the other 45 cases the VC only commits to the minimum required lock-up period of six months. The empirical results are summarized in Table 5. We observe a higher underpricing (45.7%) if the VC commits only to the minimum lock-up period. In the case of a voluntarily extended lock-up period the mean underpricing is only half the size (22.1%). However, contingent on the small sample size the difference in underpricing is significant only at the 10% level.

[Please insert Figure 5 and Table 5 about here]

After 6 months of trading the BHAR for venture-backed IPOs with extended lock-up periods is 67.4% (Figure 5) which is higher than the 37.1% BHAR for the IPOs with the standard 6 months period. This difference, however, is statistically not significant. After 24 months of trading the difference increases substantially. While the BHAR of the IPOs with the longer lock-up period increases to 67.7%, the BHAR of the other venture-backed IPOs decline to -7.4%. The difference in mean returns is not statistically significant mainly due to the small sample size. Nevertheless, it appears that the venture capitalist that agrees on a longer lock-up period signals the quality of the IPO. Whether the venture capitalist actually exits at the end of

the six months periods is not clear at all because the relevant data is not available in Germany. But this seems to be the most likely scenario.

4.3.2. Bank venture-backed IPOs and lock-up commitment

In order to investigate the specific behavior of banks with equity investments in IPOs with respect to lock-up periods and exit strategies we separate the 44 bank venture-backed IPOs by the length of the agreed lock-up period. As already mentioned in the previous section an extended lock-up period could signal IPO quality. Therefore, one could expect a positive and superior long-run performance. This may be especially true given the private information advantage of banks in general. The empirical results are presented in Table 6 and Figure 6. One IPO has to be excluded because the bank sold all shares at the time of the IPO.

[Please insert Figure 6 and Table 6 about here]

The empirical results are quite different from the ones of the non-bank venture-backed IPOs as presented in Table 4. We do not find any difference in mean underpricing for bank-financed IPOs related to the length of the lock-up commitment. The stock market performance (BHAR) for bank-financed IPOs with extended lock-up periods is negative and reaches a level of -9.0% after 6 months. This figure is not only negative but significantly lower than the performance of the other bank-financed IPOs. In contrast, they generate a positive performance of 49.5%. However, the difference in the performance of both groups after 24 months is smaller. Now the mean BHAR are 6.6% and -38.6%. For the group with lock-up agreements of only six months we observe a strong positive performance for the first six months followed by a tremendous decline after lock-up expiration. It appears that a longer lock-up period of bank venture-backed IPOs are not a credible signal to convey the quality of the IPO to the market. In this instance the extended lock-up agreement is irrelevant and has little value.

4.4. Exit in the primary market

The performance of the venture-backed IPOs during the first 6 months of trading suggests clearly that it is not sensible for the venture capital firm to exit in the primary market at the time of the going public. One obvious reason is the substantial initial return. Underpricing has

been observed for most IPOs in basically all equity markets and some underpricing is justified on economic grounds. In the case of positive initial returns the venture capital firm would leave some money on the table by exiting at the time of the IPO (Loughran and Ritter, 2002). Nevertheless, it is still interesting to investigate empirically whether the venture capitalist indeed sells any of his shares at the time of the public offering and whether this behavior can be used as an indicator for the future performance of the IPO. We therefore separate our sample into two groups depending on the fraction of share ownership that is sold in the primary market. As in other studies we use the median value of shares sold at a cut off rate (Brav and Gompers, 2003). Again, we first examine venture-backed IPOs (4.4.1) and then bank-financed IPOs (4.4.2).

4.4.1. *Exit of venture capitalists in the primary market*

If the VC sells more shares than the median value at the time of the IPO, the mean underpricing (54.4%) is significantly higher than for the other venture-backed IPOs (27.4%) with a high retention rate in ownership (Table 7, Figure 7). Obviously the investors anticipate this possible conflict of interest and require a higher initial return due to the VC's early exit. However, we do not find any significant difference in performance after 6 months of trading (39.3% and 47.4%, respectively). For both sub-samples, shares usually can be sold on average at a relatively higher price at the end of the lock-up period. Thus, venture capitalists who exit at the time of the IPO leave on average money on the table first due to the underpricing or initial returns and second due to the superior performance during the first 6 months of trading. After 24 months the BHAR are -0.3% for the IPOs without substantial exits in the primary market and 41.1% for the rest of the venture-backed IPOs. Thus, there is a strong decline in BHAR after 6 months if VCs keep their shares at the IPO date. However, the difference in mean BHAR is not statistically significant. Thus, the fact that the venture capitalist does not exit immediately at the time of the IPO does not serve as a positive signal for the market in the long-run (24 months). It may be fair to assume that this finding is not surprising and mainly due to a major exit of the VC after 6 months of trading.

[Please insert Figure 7 and Table 7 about here]

4.4.2. *Exit of banks in the primary market*

From Table 4 it is evident that bank venture-backed IPOs have a positive performance for the first 6 months of 26.4%. If we include the first day returns of 73.6% of this sample in calculating the total performance, it becomes quite obvious that a bank exit at the time of the IPO in the primary market was on average not an optimal exit strategy. Therefore, it is of interest whether the group of IPOs in which the banks sold their shares immediately in the primary market has an inferior performance compared to the group of IPOs in which the banks kept their investment at least up to the end of the lock-up period. The basic idea is that an equity commitment by a bank of a time period of at least to the end of the lock-up period signals the belief that these are higher quality firms. Consequently, a superior performance of these IPOs could be expected. The percentage of ownership that must be sold to fall into one of the two groups is again the median value of shares sold at the IPO date. The empirical results are presented in Table 8 and Figure 8.

[Please insert Figure 8 and Table 8 about here]

Mean underpricing for bank-financed IPOs is slightly higher if banks sell more shares than the median value at the time of the IPO (87.8% versus 59.4%). The difference in underpricing is not statistically significant but the performance of the two groups over the first 6 months is quite different. The group of IPOs in which banks sold immediately more secondary shares has a significant negative performance of -17.4% after 6 months whereas the group of IPOs in which the banks kept their investments has a significant positive performance of 70.3%. The difference is significant. Obviously, a strategy in which the banks keep their equity stake for a while and sold it in the secondary market yields a much better performance. The difference in performance, however, declines later on and this difference is not statistically significant after 24 months.

Whether the banks in fact used this apparent profit opportunity to exit at the end of the lock-up period cannot be investigated due to the unavailability of the relevant data. The same is true for other investors that may bandwagon on this price behavior. Depending on whether the

investor got shares allocated at the time of the IPO or bought shares in the secondary market immediately following the IPO, the investors may be able to generate on average some superior returns. Nevertheless, the cause and effect of this relationship with respect to the return behavior is not that clear at all and provides some room for speculation. From the empirical evidence it is obvious that the banks acting in their own interest should keep the shares up to the end of the 6 months period and then should start liquidating their equity stake. On the other hand, it could be argued that the positive performance at the end of the lock-up period is not random but in fact a result of the banks' objective and activities. Possible explanations for such a price behavior are price support and positive analyst recommendations with the objective to create a favorable exit environment for their own investments. Consequently, the BHAR after 6 months can be interpreted as an intended overvaluation that vanishes slowly during the next 18 months. It is needless to say that such a bank strategy in the securities market is very problematic because it would disadvantage investors that invest in shares of previous IPOs at that time. In the next section we will gather some additional evidence on the bank behavior in the German universal banking system.

4.5. Analyst recommendations

Buy and sell recommendations by analysts usually have an impact on stock prices at least in the short run (see Stickel, 1995, Desai et al., 2000, and Barber et al., 2001). Following some recent scandals in relation to IPOs in the United States the independence of analysts have been questioned and new regulations have been implemented by the SEC. Of interest are now analyst recommendations of IPOs at the Neuer Markt in which banks provided venture capital financing before the time of the IPO. Thus, banks acted as an analyst and venture capitalist for these IPOs. Analyst recommendations were collected from www.aktiencheck.de. For 2 IPOs analyst recommendations were not available. These IPOs are excluded so that there remain 42 bank-financed IPOs in this sub-sample. Altogether we find 50 analyst recommendations from banks which had equity stakes in an IPO during the first 12 months of IPO trading. There are three recommendations that could be labeled as neutral. In 47 instances the analysts of the banks recommended to buy the shares. Obviously, it is not surprising at all that the banks usually do

not recommend to their clients to divest the shares of IPOs in which they hold themselves an equity stake, especially when they cannot sell their own stake before the end of the lock-up period. However, the number of buy recommendations for an IPO is quite different. For 14 out of the 42 IPOs we observe at least two buy recommendations from the invested bank. In these instances it may be assumed that the objective of the bank is to positively influence the price performance of the IPO. In Figure 9 and Table 9 the performance of these 14 IPOs is compared to the performance of the 28 IPOs that have received none or only one buy recommendation from the analysts of the underwriting and equity holding bank. Due to the small sample size the statistical tests and therefore the results have to be interpreted very carefully.

[Please insert Figure 9 and Table 9 about here]

The IPOs for which the invested bank issued at least two buy recommendations have a significant BHAR of 86.6% over the first 6 months period. This is a much better performance than the insignificant -0.7% of the other bank-financed IPOs. The difference is statistically significant at the 10% level. After 24 months of trading, however, the difference in returns has diminished again substantially. There are different ways in which these results can be interpreted. The empirical results suggest to some extent that this is a situation with tremendous potential for conflicts of interest between banks and investors. Banks are in a position to take advantage of private information and they can be actively engaged in influencing stock prices in order to generate excess returns for their own portfolio. Overall, there appears to be some evidence that the banks used their privilege to issue buy recommendations in order to first support the stock prices and then to exit afterwards at very favorable prices. Obviously, these recommendations are not in the interest of the ordinary investor. Thus, from the perspective of the small investor the equity ownership of banks is very problematic especially when the bank also acts as an underwriter or is part of the underwriter group.

There are a variety of obvious conflicts of interests that are real and not academic when a bank acts as equity holder, underwriter, designated sponsor, analyst as well as commercial banker. From this perspective and with the empirical evidence provided in this study it is quite obvious that at least in the U.S. there are currently some institutional changes discussed and

implemented that are in the interest of the ordinary investor and that are useful in reducing potential agency problems of the investment banking business (NASD Rule 2711 and NYSE Rule 472). Because these problems are even more dramatic in a universal banking system the German authorities need to act in order to reduce the apparent conflicts of interest for banks and provide a fair and equal equity market.

5. Regression analysis

In order to check for the robustness and validity of our empirical results with respect to underpricing and long-run performance (6 and 24 months of trading) of venture-backed and non venture-backed IPOs in Germany we perform some additional regression analyses. The independent variables include the control variables “firm size” [ln firm size] and “market-to-book-ratio” [ln market/book]. Firm size corresponds with the market value of equity. These two variables adjust the IPO performance for the well known impact of size and growth potential (Lyon et al., 1999, see also Gompers and Lerner, 1999b, Espenlaub et al., 1999). The firm-size-ratio and market-to-book-ratio are based upon the first market price of the IPO. We test 7 different equations (model 1 to 7). In model 1 to 4 the following 5 dummy variables are included:

- venture-backed in the very general sense (including bank ownership) [Venture Capital]
- venture-backed in the narrow sense (excluding bank ownership) [VC Non-bank]
- venture-backed by a bank [VC Bank]
- venture-backed by a firm belonging to the underwriter consortium [VC UW Syndicate]
- venture-backed by a firm that is also the lead underwriter [VC Lead UW]

The sample sizes in the regressions for model 1 to model 4 are 307 IPOs. In model 1 we examine the impact of venture capital in general whereas in model 2 we control for the fact whether the IPO is financed by an independent VC or by a bank subsidiary. In some cases the IPO was financed by an independent VC and a VC bank so that both variables do not exclude each other. Model 3 and 4 investigate the impact of VC underwriter affiliation on IPO performance. In model 5 the sample sizes decreases to 101 due to the fact that only venture-backed IPOs in the general sense are analyzed with respect to their pre IPO ownership, exit behavior, and lock-up commitment. In model 6 and 7, the same methodology is applied for independent VCs

and bank affiliated VCs, respectively. Thus, in model 5 to 7, we distinguish between three different groups: The first group includes all 101 IPOs with venture capital-financing (model 5), the second group includes those 72 IPOs that are backed by non-bank venture capitalists (model 6), and the third group includes 44 bank venture-backed IPOs (model 7). For all three groups we include

- ownership structure before the IPO [VC, VC Non-bank, VC Bank Ownership]
- change in ownership at the time of the IPO, i.e., the selling of shares in the going public process, where a sell has a negative sign [Δ VC, Δ VC Non-bank, Δ VC Bank]
- length of the agreed lock-up period [VC, VC Non-bank, VC Bank Lock-up].

The results are presented in Tables 10 to 12.

5.1. *IPO underpricing*

The empirical findings for underpricing are presented in Table 10. First of all both control variables are significant indicating a positive size effect and a positive market-to-book ratio effect. Similar results are reported in a study by Tykvova and Walz (2004). The most important finding is that all coefficients in model 2, 3, and 4 are significant. In these models we investigate the conflict of interest of banks by testing for the impact of bank venture capital on underpricing. The additional variables tested do not have a significant impact at least not on underpricing (models 5, 6, 7). These empirical findings are now discussed in more detail.

The results of model 1 suggest that venture-backed IPOs have a slightly but insignificantly higher underpricing than non venture-backed IPOs. This finding is at first surprising and is in contrast to the certification hypothesis of venture capital involvement, but these results are in line with the findings by Franzke (2001) for Germany and Francis and Hasan (2001) for the United States. Obviously, VCs in general do not contribute to reduce IPO underpricing. The results of model 2, however, offer additional insights and reveal that the bank venture-backed IPOs have significantly higher initial returns. In contrast, the non-bank venture capitalists have a slightly negative but insignificant impact on underpricing. These findings are in line with the results reported in Table 4 and support the view, that the higher underpricing of bank-financed IPOs is the consequence of possible conflicts of interests in a universal banking system. Thus,

investors require high initial returns for IPOs in the primary market to compensate them for the additional risk. Another explanation of this result could be that banks set the issue price too low. Reasons for such a behavior could be incentive problems (Baron, 1982) or allocation arguments, i.e., banks could allocate underpriced shares to special investors or clients (spinning activities as reported in Loughran and Ritter, 2003). The most interesting aspect of our results is that the underpricing is significant when the venture capitalist belongs to the underwriter consortium (model 3) or even acts as the lead underwriter (model 4). Again, the positive coefficients are evidence of some conflicts of interest. In conclusion, these empirical findings support the idea that there are conflicts of interests in venture-backed IPOs when the venture capital provider is not independent and most likely follows his own agenda for maximizing his return objectives, i.e., issuing overvalued IPOs.

In addition to the impact of venture capital, there are other factors that may contain important information that could be reflected in the magnitude of the underpricing such as the level of VC ownership before the IPO, the change in ownership structure at the time of the IPO as well as an extended lock-up period of the venture capitalist. These variables are tested in model 5, 6, and 7. Although the adjusted R^2 is above 0.50 in these models, the coefficients for these variables are not statistically significant indicating that there is no impact on underpricing. However, the size and market-to-book ratios are still significant. Nevertheless, the empirical findings in this section provide evidence of conflicts of interests in the going public process when a bank is involved as a venture capitalist and is part of the underwriter syndicate or is the lead underwriter.

5.2. Abnormal returns after 6 months of trading

In most cases the lock-up period for venture-backed IPOs ends after the legally required period of 6 months of trading and the venture capitalists have the opportunity to sell the rest of their shares in the secondary market. Thus, the IPO performance after 6 months offers some interesting insights into the exit behavior. The regression results are documented in Table 11. In most models the control variable firm size is not significant whereas the “market-to-book” ratio has a significantly negative impact on the BHAR, meaning that IPOs with higher expected

growth rates have an inferior performance in the secondary market relative to the other IPOs. This suggests that the expectations and valuation at the time of the IPO were too optimistic. The results support the common knowledge that it is not a good strategy to overpay for expected growth (Haugen, 2001, p.158). The dummy variables in model 1 to 4 do not have any significant impact on the BHAR after 6 months. For the broad definition of venture capital (model 5) we find that the length of the lock-up period has a negative and significant impact on the performance.

The empirical findings for bank venture-backed IPOs offer the most interesting insights (model 7). When the bank has a higher percentage of ownership and sells more shares at the time of the IPO we observe an inferior performance during the first 6 months. This suggests that banks are able to time their exit in that they sell a major part of their shares in the primary market when they expect a weak stock market performance. Moreover, an extended lock-up period by banks has a negative impact, indicating that a longer commitment cannot serve as a credible signal. Thus, banks are not able to produce a positive signal in case of a weak company. These results are in line with our findings documented in Tables 6 and 8. Again, this supports the hypothesis of substantial conflicts of interests especially when banks act as venture capitalist. To better understand whether an exit after 6 months represents an optimal timing we analyze the returns for the subsequent 18 months.

5.3. Abnormal returns after 24 months of trading

The regression results documented in Table 12 reveal that after 24 months of trading the independent variables do not explain any variation of the performance in these models. We find only some significance in model 7 in which a longer lock-up period by banks has a negative impact on the returns. This suggests again that an extended lock-up period by banks is not viewed as a positive and convincing signal by the market. Because the positive valuation effects often lasts only for the first 6 months and are in general followed by a substantial underperformance over the subsequent 18 months this return behavior suggests a possible temporary overvaluation. This offers very favorable exit opportunities for the venture capitalist at that time. It needs to be recognized, however, that a very favorable exit timing of the venture capitalist may

be based on private information and means that the investors that bought these shares in the secondary market did not get a very fair deal. Unfortunately, there is no reliable data base that reveals exactly when the venture capitalist exited the venture. The empirical results, however, support the notion that the venture capitalist sold his shares at the assumed date.

It is therefore important for the financial market in Germany that some reliable data is available showing the exit behavior of important investor groups. This would also mean more investor protection. In 2002, these facts became so obvious that the German government extended the disclosure requirements for the change in ownership concentration in that the threshold for disclosure was lowered from 25% to 5%. Unfortunately, there are still no special disclosure requirements for major pre IPO shareholders in Germany. In sum, the legal environment for public shareholders improves slowly but without reaching an international comparable and appropriate level.

6. Conclusions

The objective of this study was to investigate the financial performance of non-bank and bank venture-backed IPOs at the “Neuer Markt” in Germany for the period from 1998 to 2001. Venture capital firms and possibly banks are usually in a position to signal reliably the superior quality of an IPO (certification hypotheses), thereby reducing agency problems. This should result in a smaller underpricing and a better long-run performance. Various theoretical and empirical studies suggest, however, that there are substantial conflicts of interest if a commercial banker acts simultaneously as venture capitalist, underwriter, and provides investment research.

The empirical findings of our research suggest that there are in fact serious agency problems in the going public process, especially when banks act as venture capitalist and underwriter. These IPOs are significantly more underpriced. Because pre IPO shareholders are locked-in for the first six months of trading, venture-backed IPOs with the standard six months lock-up period have a strong out-performance during the first six months of trading and a significant underperformance thereafter. In cases of longer lock-up periods, e.g., 12 or 18 months,

the initial outperformance continues but does not increase any further. For bank venture-backed IPOs we observe the opposite results. Extended lock-up periods of banks convey a negative signal suggesting that the investor is aware of these problems. If the lock-up period, for example, is only six months, most banks usually find ideal opportunities for an optimal exit. Considering the underpricing and long-run performance of the IPOs at the “Neuer Markt”, venture capitalists and banks would have left on average money on the table if they sold large equity stakes at the time of the initial public offering.

The empirical results of this study support the view that a substantial sale of primary shares by banks and venture capitalists at the time of the IPO should be interpreted as a negative signal. It appears that especially banks are able to time their exit quite perfectly. The interesting question, however, is whether banks have certain timing abilities or whether they use private information from their equity stake (venture capital), the due diligence process (underwriting) or employ other means to influence the stock price (analyst recommendations). The empirical results support the idea that banks manage to some extent the stock prices of IPOs by using over-optimistic analyst recommendations. In conclusion, there seems to be sufficient empirical evidence that in the German universal banking system financial institutions act in their own interest or in the interest of specific investor groups but to the disadvantage of the ordinary investor.

REFERENCES

- Ang, J. S. and Richardson, T., 'The Underwriting Experience of Commercial Bank Affiliates prior to the Glass-Steagal Act: A Reexamination of Evidence for Passage of the Act', *Journal of Banking and Finance*, Vol. 18, 1994, pp. 351-395.
- Barber, B., Lehavy, R., McNichols, M. and Trueman, B., 'Can Investors Profit from the Prophets? Security Analyst Recommendations and Stock Returns', *Journal of Finance*, Vol. 66, 2001, pp. 531-563.
- Baron, D. P., 'A Model of the Demand for Investment Banking Advising and Distribution Services for New Issues', *Journal of Finance*, Vol. 37, 1982, pp. 955-976.
- Barry, C. B., Muscarella, C. J., Peavy III, J. W. and Vetsuypens, M. R., 'The Role of Venture Capital in the Creation of Public Companies', *Journal of Financial Economics*, Vol. 27, 1990, pp. 447-471.
- Baums, T. and Theissen, E., 'Banken, bankeigene Kapitalanlagegesellschaften und Aktienemissionen', *Zeitschrift für Bankrecht und Bankwirtschaft*, Vol. 11, 1999, pp. 125-204.
- Ber, H., Yafeh, Y. and Yosha, O., 'Conflict of Interest in Universal Banking: Bank Lending, Stock Underwriting, and Fund Management', *Journal of Monetary Economics*, Vol. 47, 2001, pp. 189-218.
- Bessler, W. and Thies, S., 'The Long-Run Performance of Initial Public Offerings in Germany', *Working Paper* (University of Giessen, 2004).
- Brav, A. and Gompers, P. A., 'Myth or Reality? The Long-Run Underperformance of Initial Public Offerings: Evidence from Venture and Nonventure Capital-Backed Companies', *Journal of Finance*, Vol. 52, 1997, pp. 1791-1821.
- Brav, A. and Gompers, P. A., 'The Role of Lock-ups in Initial Public Offerings', *Review of Financial Studies*, Vol. 16, 2003, pp. 1-29.
- Desai, H., Liang, B. and Singh, A. K., 'Do All-Stars Shine? Evaluation of Analyst Recommendations', *Financial Analysts Journal*, Vol. 56, May/June 2002, pp. 20-29.
- Drobetz, W., Kammermann, M. and Wälchli, U., 'Performance of Initial Public Offerings: The Evidence for Switzerland', *Working Paper* (Universities of Basel and Bern, 2003).
- Espenlaub, S., Garrett, I. and Mung, W. P., 'Conflicts of Interest and the Performance of Venture-Capital-Backed IPOs: A Preliminary Look at the UK', *Venture Capital*, Vol. 1, 1999, pp. 325-349.
- Espenlaub, S., Goergen, M. and Khurshed, A., 'IPO Lock-in Agreements in the UK', *Journal of Business Finance & Accounting*, Vol. 28, 2001, pp. 1235-1278.
- Field, L. C. and Hanka, G., 'The Expiration of IPO Share Lockups', *Journal of Finance*, Vol. 56, 2001, pp. 471-500.
- Fields, P., Fraser, D. R. and Bhargava, R., 'A Comparison of Underwriting Costs of Initial Public Offerings by Investment and Commercial Banks', *Journal of Financial Research*, Vol. 26, 2003, pp. 517-534.
- Fields, P. and Fraser, D. R., 'Effects of IPO Mispricing on the Risk and Reputational Capital of Commercial Banks', *Review of Financial Economics*, Vol. 13, 2004, pp. 65-77.
- Francis, B. B. and Hasan, I., 'The Underpricing of Venture and Nonventure Capital IPOs: An Empirical Investigation', *Journal of Financial Services Research*, Vol. 19, 2001, pp. 99-113.
- Franzke, S. A., 'Underpricing of Venture-Backed and Non Venture-Backed IPOs: Germany's Neuer Markt', *Working Paper* (University of Frankfurt, 2001).
- Franzke, S. A. and Schlag, C., 'Over-Allotment Options in IPOs in Germany's Neuer Markt: An Empirical Investigation', *Working Paper* (University of Frankfurt, 2002).

- Giudici, G. and Roosenboom, P., 'Pricing Initial Public Offerings in Europe: What Has Changed?', *Working Paper* (University of Bergamo, 2002).
- Goergen, M., Renneboog, L. and Khurshed, A., 'Lock-in Agreement in French Nouveau Marché and German Neuer Markt IPOs', *Working Paper* (Manchester School of Management, 2004).
- Gompers, P. A., 'Grandstanding in the Venture Capital Industry', *Journal of Financial Economics*, Vol. 42, 1996, pp. 133-156.
- Gompers, P. A. and Lerner, J., 'Venture Capitalist Distributions: Short-Run and Long-Run Reactions', *Journal of Finance*, Vol. 53, 1998, pp. 2161-2183.
- Gompers, P. A. and Lerner, J., *The Venture Capital Cycle* (MIT Press: Cambridge, Mass., 1999a).
- Gompers, P. A. and Lerner, J., 'Conflict of Interest in the Issuance of Public Securities: Evidence from Venture Capital', *Journal of Law and Economics*, Vol. 42, 1999b, pp. 1-28.
- Hamao, Y., Packer, F. and Ritter, J. R., 'Institutional Affiliation and the Role of Venture Capital: Evidence from Initial Public Offerings in Japan', *Pacific-Basin Finance Journal*, Vol. 8, 2000, pp. 529-558.
- Haugen, R. A., *Modern Investment Theory*, 5th edition (Prentice Hall: New York, 2001).
- Hebb, G. M., 'Conflict of Interest in Commercial Bank Equity Underwriting', *Financial Review*, Vol. 37, 2002, pp. 185-206.
- Kiss, I. and Stehle, R., 'Underpricing and Long-Term Performance of Initial Public Offerings at Germany's Neuer Markt', *Working Paper* (Humboldt University Berlin, 2002).
- Kraus, T. and Burghof, H.-P., 'Post-IPO Performance and the Exit of Venture Capitalists', *Working Paper* (University of Munich, 2003).
- Kroszner, R. S. and Rajan, R. G., 'Is the Glass-Steagall Act Justified? A Study of the U.S. Experience with Universal Banking Before 1933', *American Economic Review*, Vol. 84, 1994, pp. 810-832.
- Lee, P. M. and Wahal, S., 'Grandstanding, Certification and the Underpricing of Venture Capital Backed IPOs', *Journal of Financial Economics*, Vol. 73, 2004, pp. 375-407.
- Lerner, J., 'Venture Capitalists and the Decision to Go Public', *Journal of Financial Economics*, Vol. 35, 1994, pp. 293-316.
- Li, X. and Masulis, R. W., 'Venture Capital Investments by IPO Underwriters: Certification or Conflict of Interest?', *Working Paper* (University of Miami and Vanderbilt University, 2003).
- Lin, H.-W. and McNichols, M. F., 'Underwriting Relationship, Analysts' Earnings Forecasts and Investment Recommendations', *Journal of Accounting and Economics*, Vol. 25, 1998, pp. 101-127.
- Ljungqvist, A. P. and Wilhelm, W. J., 'What makes Issuers Happy? Testing the Prospect Theory of IPO Underpricing', *Working Paper* (New York University and University of Virginia, 2003).
- Löffler, G., 'Anatomy of a Performance Race', *Working Paper* (University of Ulm, 2003).
- Loughran, T. and Ritter, J. R., 'Why Don't Issuers Get Upset About Leaving Money on the Table in IPOs?', *Review of Financial Studies*, Vol. 15, 2002, pp. 413-443.
- Loughran, T. and Ritter, J. R., 'Why Has IPO Underpricing Changed over Time', *Working Paper* (Universities of Notre Dame and Florida, 2003).
- Lyon, J. D., Barber, B. M. and Tsai, C.-L., 'Improved Methods for Tests of Long-Run Abnormal Stock Returns', *Journal of Finance*, Vol. 54, 1999, pp. 165-201.

- Meggison, W. L. and Weiss, K. A., 'Venture Capitalist Certification in Initial Public Offerings', *Journal of Finance*, Vol. 46, 1991, pp. 879-903.
- Michaely, R. and Womack, K. L., 'Conflict of Interest and the Credibility of Underwriter Analyst Recommendation', *Review of Financial Studies*, Vol. 12, 1999, pp. 653-686.
- Nowak, E. and Gropp, A., 'Ist der Ablauf der Lock-up-Frist bei Neuemissionen ein kursrelevantes Ereignis? Eine empirische Analyse von Unternehmen des Neuen Marktes', *Zeitschrift für betriebswirtschaftliche Forschung*, Vol. 54, 2002, pp. 19-45.
- Puri, M., 'Commercial Banks in Investment Banking: Conflict of Interest or Certification Role?', *Journal of Financial Economics*, Vol. 40, 1996, pp. 373-401.
- Puri, M., 'Commercial Banks as Underwriters: Implications for the Going Public Process', *Journal of Financial Economics*, Vol. 54, 1999, pp. 133-163.
- Ritter, J. R., 'Differences between European and American IPO Markets', *European Financial Management*, Vol. 9, pp. 421-434.
- Schenone, C., 'The Effect of Banking Relationships on the Firm's IPO Underpricing', *Journal of Finance*, 2004, forthcoming.
- Stickel, S. E., 'The Anatomy of the Performance of Buy and Sell Recommendations', *Financial Analysts Journal*, Vol. 51, September/October 1995, pp. 25-39.
- Tyková, T. and Walz, U., 'Are IPOs of Different VCs Different?', *Working Paper* (Universities of Mannheim and Frankfurt, 2004).
- Ursel, N. D. and Ljucovic, P., 'The Impact of Bank Ownership of Underwriters on the Underpricing of Initial Public Offerings', *Canadian Journal of Administrative Sciences*, Vol. 15, 1998, pp. 17-27.

Figures and Tables

Figure 1:
Outline of the empirical study

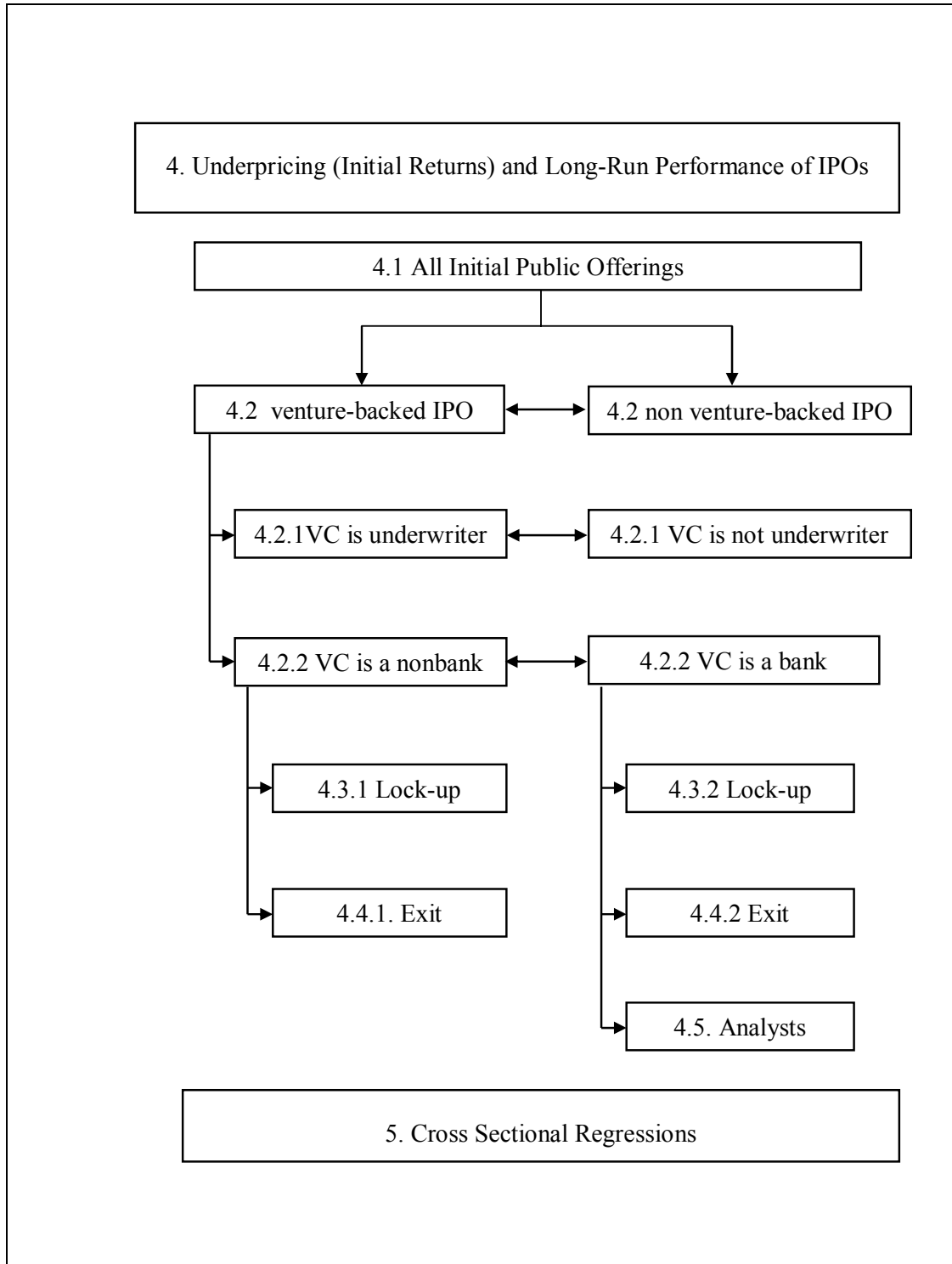


Table 1
Descriptive statistics

This table contains a description of our data. In Panel A, we apply a broad definition of VC ownership. Venture-backed are those IPOs, where a venture capital firm (included bank ownership) is involved in an IPO at the date of the going public and owns more than 4% of shareholdings pre IPO. In Panel B, we exclude bank ownership. Thus, independent VC own more than 4% of IPO stock. Panel C includes 44 bank-financed IPOs. Again, we apply a cut-off rate of 4% ownership. Lock-up and underwriter affiliation data are taken from each IPO prospectus. The asterisk (*) indicates that the shareholder sold all shares at the IPO date.

	Mean	Median	STD	Max.	Min.
Panel A: Venture capital (non-banks & banks, 101 IPOs)					
VC ownership pre IPO	28.34%	22.35%	20.28%	89.8%	4.29%
VC ownership post IPO	15.96%	12.85%	12.29%	71.3%	0,00%
Percentage of shares sold in the IPO	24.91%	21.96%	24.1%	100,00%	-49.74%
Lock-up commitment (in months)	7.72	6	3.61	30	0*
No. of IPOs with extended lock-up	27				
No. of IPOs VC is underwriter	35				
No. of IPOs VC is lead underwriter	22				
Panel B: Venture capitalist is non-bank (72 IPOs)					
VC non-bank ownership pre IPO	29.21%	23.3%	21.74%	89.8%	4.4%
VC non-bank ownership post IPO	16.43%	12.68%	13.29%	71.3%	1.02%
Percentage of shares sold in the IPO	24.22%	22.53%	20.78%	89.1%	-3.74%
Lock-up commitment (in months)	7.29	6	2.46	12	6
No. of IPOs with extended lock-up	16				
Panel C: Venture capitalist is bank (44 IPOs)					
VC bank ownership pre IPO	16.83%	13.77%	10.83%	44.67%	4.29%
VC bank ownership post IPO	9.34%	8.05%	6.61%	24.52%	0,00%
Percentage of shares sold in the IPO	27.88%	21.13%	27.58%	100%	-4.1%
Lock-up commitment (in months)	8.45	6	4.68	30	0*
No. of IPOs with extended lock-up	14				

Figure 2
Venture-backed versus non venture-backed IPOs

The figure shows the long-run performance (BHAR) of venture-backed and non venture-backed IPOs. The firms are divided into two sub-samples. Venture-backed are those IPOs, where a venture capital firm (included bank ownership) is involved in an IPO at the date of the going public. The remaining IPOs are defined as non venture-backed. 123 trading days correspond to 6 months and 500 trading days to 24 months of trading in the secondary market. The end of the lock-up period is shown by the vertical line.

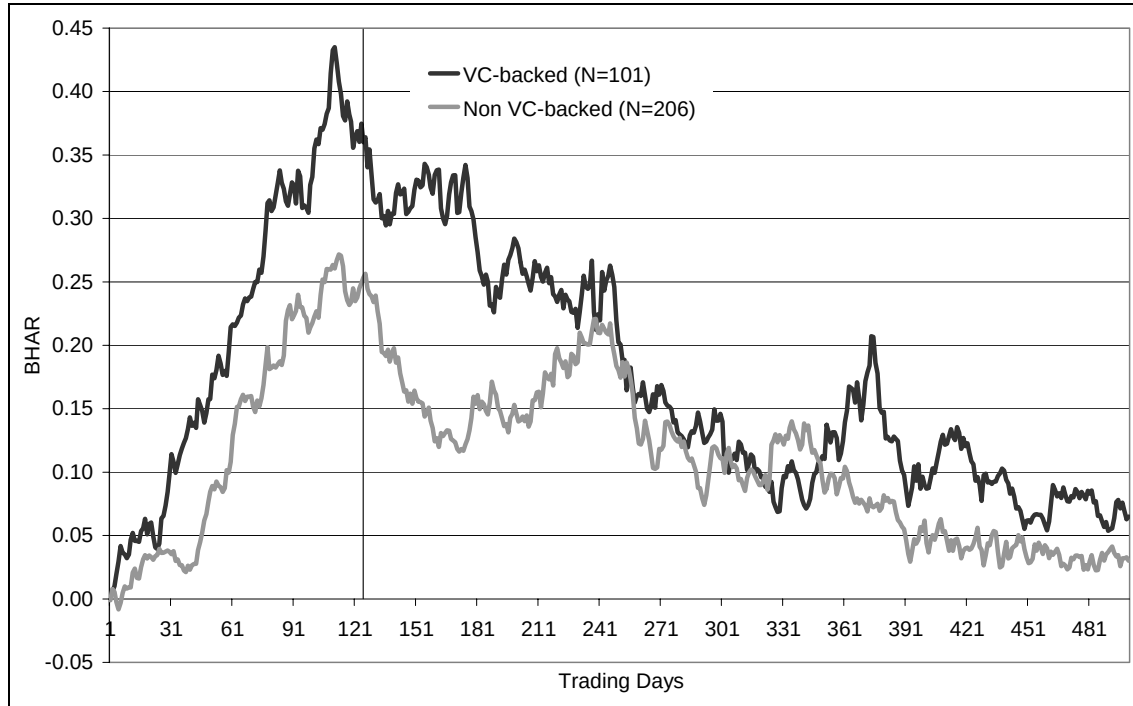


Table 2
Venture-backed versus non venture-backed IPOs

This table contains the mean underpricing and buy and hold abnormal return (BHAR) of venture-backed (VB) and non venture-backed IPOs (NVB) after 123 and 500 days of trading (about 6 and 24 months, respectively). To test whether the BHAR are significantly different from zero we apply Student's skewness-adjusted t-test (t adj.) and a simple sign test (z Sign). Furthermore we test whether the mean returns of both sub-samples differ significantly applying a t-test for differences in means (t means) and Wilcoxon's test of sum of ranks (U-Test). a/b/c denotes significance at the 1%/ 5% /10% level.

	N	Mean	Median	% Neg.	Std. Dev.	Skew.	Kurt.	t (adj.)	Z (Sign)	t (means)	z (U-Test)
Underpricing											
VB IPO	101	55.0%	19.0%	8.9%	0.76	1.79	6.27	10,34 ^a	8,36 ^a	0.87	0.70
NVB IPO	206	47.1%	17.5%	11.2%	0.72	2.17	8.46	13,77 ^a	11,22 ^a		
123 Trading Days											
VB IPO	101	36.0%	4.1%	48.5%	1.08	2.46	10.23	4,29 ^a	0.40	0.87	1.16
NVB IPO	206	24.5%	-4.8%	54.9%	1.13	2.91	14.41	3,77 ^a	-1.32		
500 Trading Days											
VB IPO	101	6.5%	-6.9%	63.4%	1.34	3.17	21.79	0.51	-2,59 ^b	0.23	0.29
NVB IPO	206	3.0%	-8.3%	65.1%	1.05	4.35	37.76	0.43	-4,25 ^a		

Figure 3
Affiliated versus non affiliated venture-backed IPOs

The figure shows the long-run performance (BHAR) of affiliated and non affiliated venture-backed IPOs. The venture-backed IPOs (N=101) are divided into two sub-samples. Affiliated are those venture-backed IPOs, where the underwriter also provided equity financing to the IPO firm and owns more than 4% of equity before going public (N=35). The rest of the venture-backed IPOs are non-affiliated (N=66). 123 trading days correspond to 6 months and 500 trading days to 24 months of trading in the secondary market. The end of the lock-up period is shown by the vertical line.

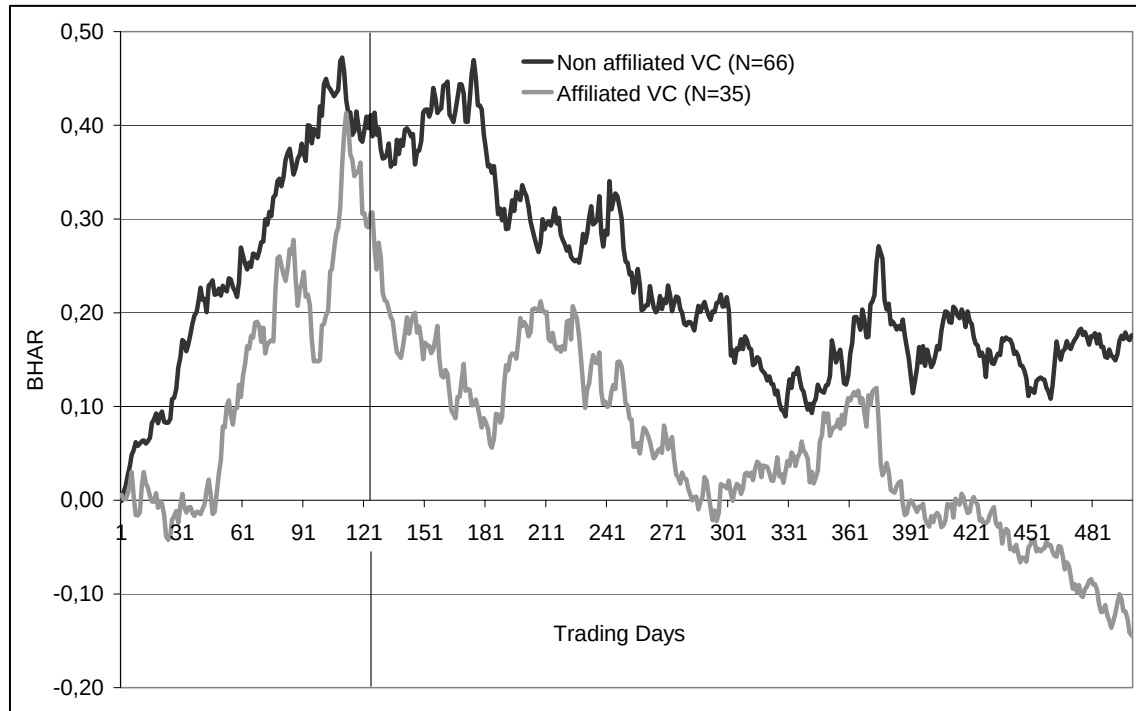


Table 3
Affiliated versus non affiliated venture-backed IPOs

This table contains the mean underpricing and buy and hold abnormal return (BHAR) of venture-backed IPOs depending on affiliation after 123 and 500 days of trading (about 6 and 24 months, respectively). To test whether the BHAR are significantly different from zero we apply Student's skewness-adjusted t-test (t adj.) and a simple sign test (z Sign). Furthermore we test whether the mean returns of both sub-samples are significantly different applying a t-test for differences in means (t means) and Wilcoxon's test of sum of ranks (U-Test). a/b/c denotes significance at the 1%/ 5% /10% level.

	N	Mean	Median	% Neg.	Std. Dev.	Skew.	Kurt.	t (adj.)	Z (Sign)	t (means)	z (U-Test)
Underpricing											
Affiliated VC	35	84.1%	61.1%	5.7%	1.00	1.21	3.78	6,71 ^a	5,41 ^a		
Non affiliated VC	66	39.6%	17.5%	10.6%	0.55	1.65	5.31	8,10 ^a	6,52 ^a	2,45 ^b	2,11 ^b
123 Trading Days											
Affiliated VC	35	29.1%	-12.0%	60.0%	1.20	2.68	10.17	1,75 ^c	-1.01		
Non affiliated VC	66	39.7%	7.9%	42.4%	1.01	2.27	10.10	4,16 ^a	1.35	-0.44	-1.35
500 Trading Days											
Affiliated VC	35	-14.5%	-7.3%	68.6%	0.92	-1.34	9.62	-0.87	-2,03 ^b		
Non affiliated VC	66	17.6%	-5.2%	60.6%	1.51	3.42	18.99	1.08	-1.60	-1.33	-0.63

Figure 4
Venture-backed versus bank-financed IPOs

The figure shows the long-run performance (BHAR) of venture capital-financed and bank-financed IPOs. The venture-backed IPOs (N=101) are divided into two sub-samples. Bank-financed are those venture-backed IPOs, where the VC is a commercial bank and owns more than 4% of equity before going public (N=44). The rest of the venture-backed IPOs are non bank-financed (N=57). 123 trading days correspond to 6 months and 500 trading days to 24 months of trading in the secondary market. The end of the lock-up period is shown by the vertical line.

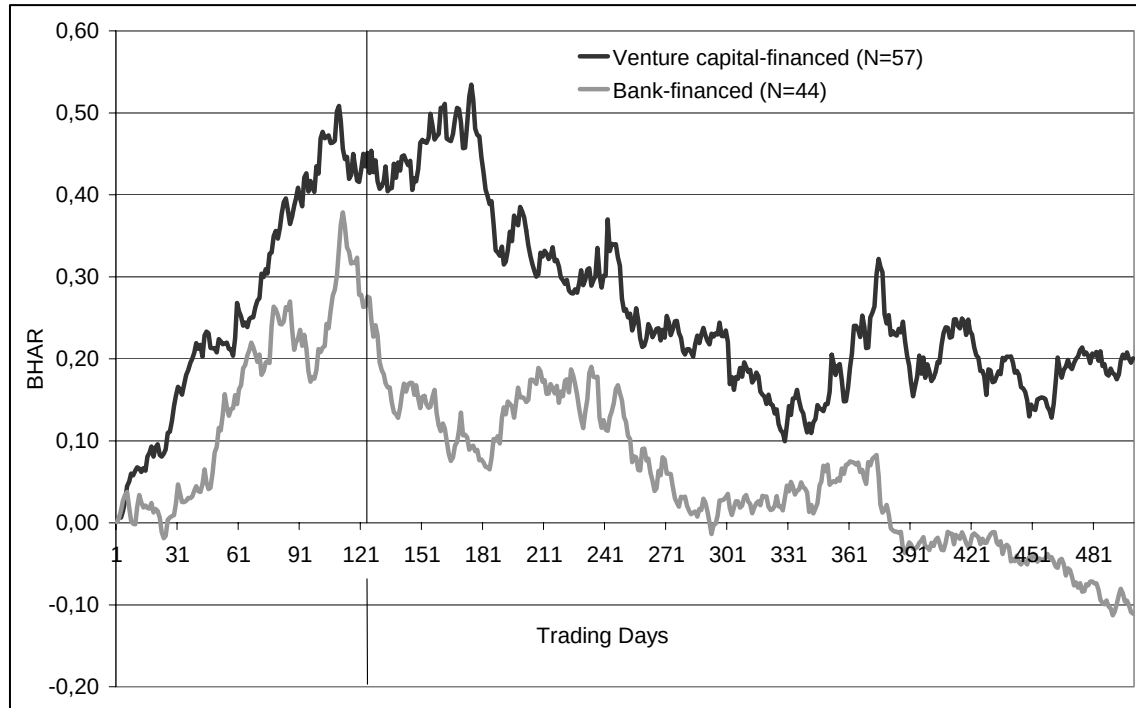


Table 4
Venture-backed versus bank-financed IPOs

This table contains the mean underpricing and buy and hold abnormal returns (BHAR) of bank-financed and venture capital-financed IPOs after 123 and 500 trading days (about 6 months and 24 months, respectively). To test whether the BHAR are significantly different from zero we apply Student's skewness-adjusted t-test (t adj.) and a simple sign test (z Sign). Furthermore we test whether the mean returns of both sub-samples are significantly different applying a t-test for differences in means (t means) and Wilcoxon's test of sum of ranks (U-Test). a/b/c denotes significance at the 1%/ 5% /10% level.

	N	Mean	Median	% Neg.	Std. Dev.	Skew.	Kurt.	t (adj.)	z (Sign)	t (means)	z (U-Test)
Underpricing											
Bank	44	73.6%	30.2%	4.5%	0.93	1.46	4.67	7,34 ^a	6,18 ^a	2,07 ^b	1,76 ^c
VC	57	40.7%	18.3%	12.3%	0.58	1.62	5.00	7,32 ^a	5,83 ^a		
123 Trading Days											
Bank	44	26.4%	-11.1%	54.5%	1.11	2.82	11.56	1,94 ^c	-0.45	-0.78	-1.39
VC	57	43.5%	11.6%	43.9%	1.06	2.20	9.43	4,05 ^a	1.06		
500 Trading Days											
Bank	44	-11.1%	-7.3%	68.2%	0.85	-1.35	10.72	-0.82	-2,26 ^b	-1.25	-0.60
VC	57	20.1%	-5.2%	59.7%	1.61	3.21	16.74	1.06	-1.32		

Figure 5
Venture-backed IPOs and the lock-up commitment

The figure shows the long-run performance (BHAR) of venture-backed IPOs without any bank ownership. The IPOs are divided into two sub-samples. In 12 cases the venture capitalists commits to a lock-up period longer than 6 months. In the remaining cases (45) the venture capitalist only commits to the minimum requirement of 6 months. 123 trading days correspond to 6 months and 500 trading days to 24 months of trading in the secondary market. The end of the lock-up period is shown by the vertical line.

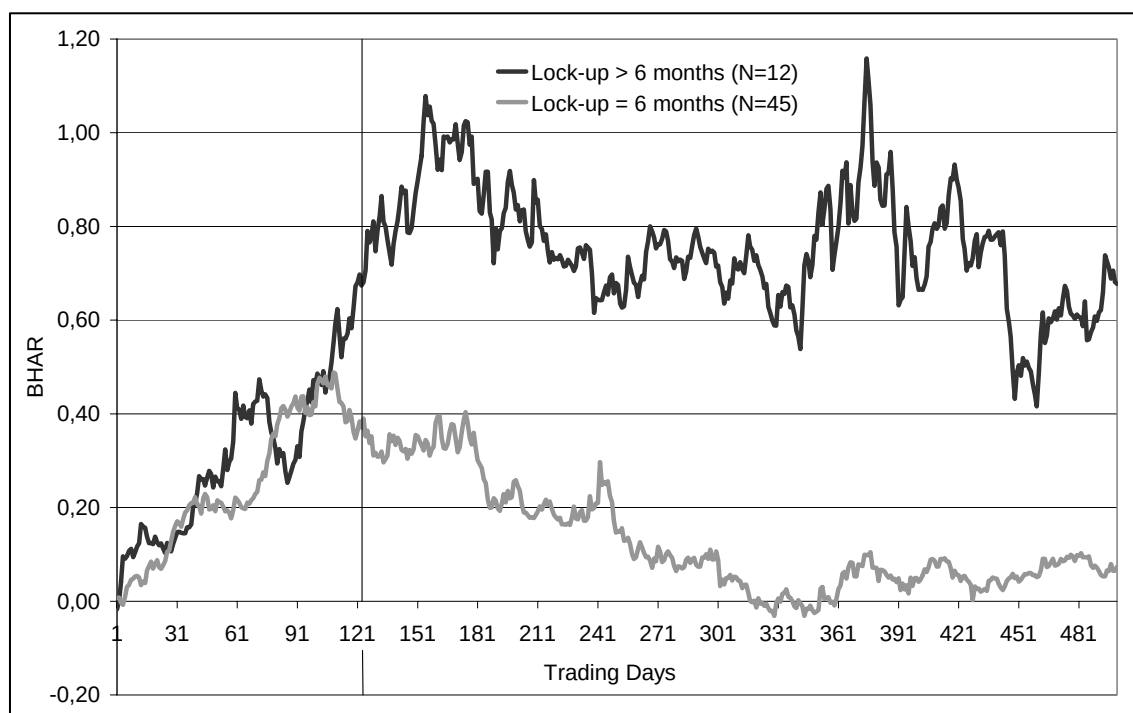


Table 5
Venture-backed IPOs and the lock-up commitment

This table contains the mean underpricing and buy and hold abnormal returns (BHAR) of venture-backed IPOs depending on its lock-up commitment after 123 and 500 trading days (about 6 months and 24 months, respectively). To test whether the BHAR are significantly different from zero we apply Student's skewness-adjusted t-test (t adj.) and a simple sign test (z Sign). Furthermore we test whether the mean returns of both sub-samples are significantly different applying a t-test for differences in means (t means) and Wilcoxon's test of sum of ranks (U-Test). a/b/c denotes significance at the 1%/ 5% /10% level.

	N	Mean	Median	% Neg.	Std. Dev.	Skew.	Kurt.	t (adj.)	z (Sign)	t (means)	z (U-Test)
Underpricing											
Lock-up = 6	45	45.7%	18.4%	15.6%	0.63	1.39	4.10	6,52 ^a	4,77 ^a		
Lock-up > 6	12	22.1%	6.0%	0.0%	0.29	1.78	5.60	3,87 ^a	3,75 ^a	1,89 ^c	0.25
123 Trading Days											
Lock-up = 6	45	37.1%	6.0%	48.9%	1.10	2.46	10.42	2,90 ^a	0.30		
Lock-up > 6	12	67.4%	37.2%	25.0%	0.86	0.63	1.96	3,20 ^a	2,02 ^b	-1.02	-1,76 ^c
500 Trading Days											
Lock-up = 6	45	7.4%	-8.0%	62.2%	1.57	3.67	21.75	0.34	-1.49		
Lock-up > 6	12	67.7%	-1.9%	50.0%	1.75	2.13	6.15	1,73 ^c	0.29	-1.08	-1.04

Figure 6
Bank-financed IPOs and the lock-up commitment

The figure shows the long-run performance (BHAR) of bank-financed IPOs. The IPOs are divided into two sub-samples. In 43 cases of the 44 bank-financed IPOs the bank still owns some shares at the IPO after the going public. In 16 cases the bank commits to a lock-up period of longer than 6 months. In the remaining cases (27) the bank only commits to the minimum requirement of 6 months. 123 trading days correspond to 6 months and 500 trading days to 24 months of trading in the secondary market. The end of the lock-up period is shown by the vertical line.

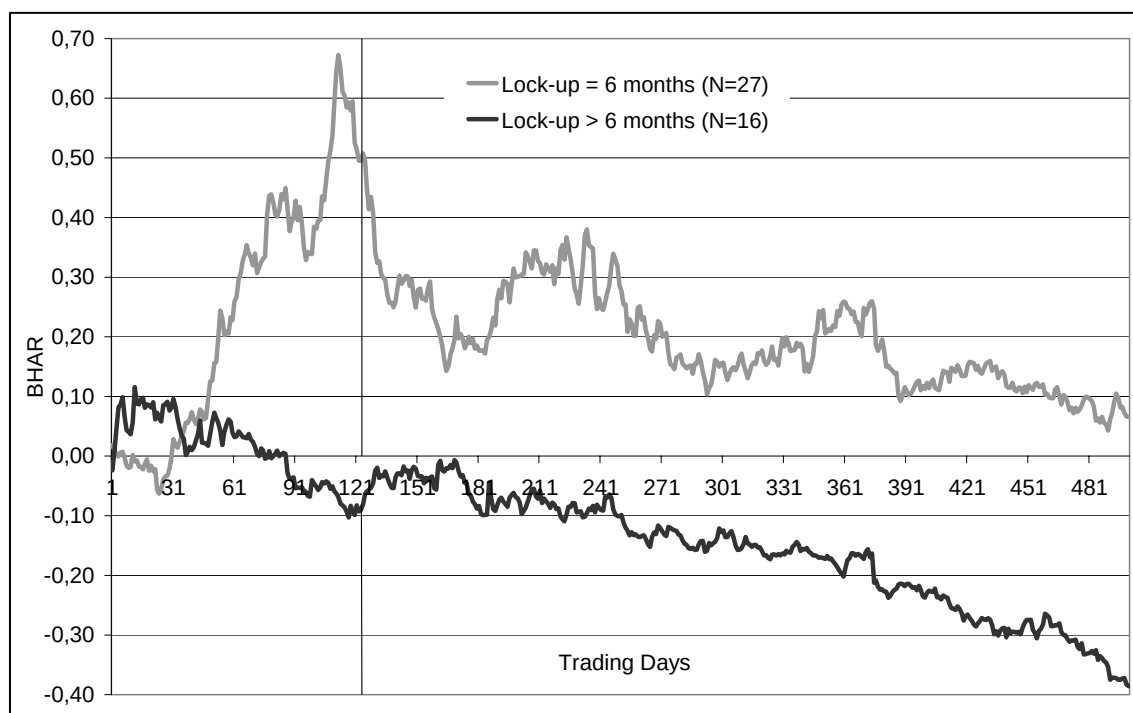


Table 6
Bank-financed IPOs and the lock-up commitment

The table reports the mean underpricing and buy and hold abnormal returns (BHAR) of bank-financed IPOs depending on its lock-up commitment after 123 and 500 trading days (about 6 months and 24 months, respectively). To test whether the BHAR are significantly different from zero we apply Student's skewness-adjusted t-test (t adj.) and a simple sign test (z Sign). Furthermore we test whether the mean returns of both sub-samples are significantly different applying a t-test for differences in means (t means) and Wilcoxon's test of sum of ranks (U-Test). a/b/c denotes significance at the 1%/ 5% /10% level.

	N	Mean	Median	% Neg.	Std. Dev.	Skew.	Kurt.	t (adj.)	z (Sign)	t (means)	z (U-Test)
Underpricing											
Lock-up = 6	27	66.7%	29.4%	3.7%	0.95	2.04	6.62	5,41 ^a	5,00 ^a	-0.26	-0.50
Lock-up > 6	16	73.9%	48.3%	6.3%	0.82	0.45	1.70	4,13 ^a	3,75 ^a		
123 Trading Days											
Lock-up = 6	27	49.5%	-10.5%	55.6%	1.35	2.08	7.06	2,40 ^b	-0.38	2,16 ^b	-1.11
Lock-up > 6	16	-9.0%	-4.4%	50.0%	0.30	-0.10	1.83	1.20	0.25		
500 Trading Days											
Lock-up = 6	27	6.6%	-6.9%	63.0%	0.75	0.71	6.12	0.47	-1.15	1.60	0.80
Lock-up > 6	16	-38.6%	-7.4%	75.0%	0.97	-2.85	10.16	1.00	-1,75 ^c		

Figure 7
Venture-backed IPOs and the sale of secondary shares in the primary market

The 57 venture-backed IPOs (without bank ownership) are separated by the criteria, whether the venture capitalist sells more or less than the median value of his shares at the primary market. 123 trading days correspond to 6 months and 500 trading days to 24 months of trading in the secondary market. The end of the lock-up period is shown by the vertical line.

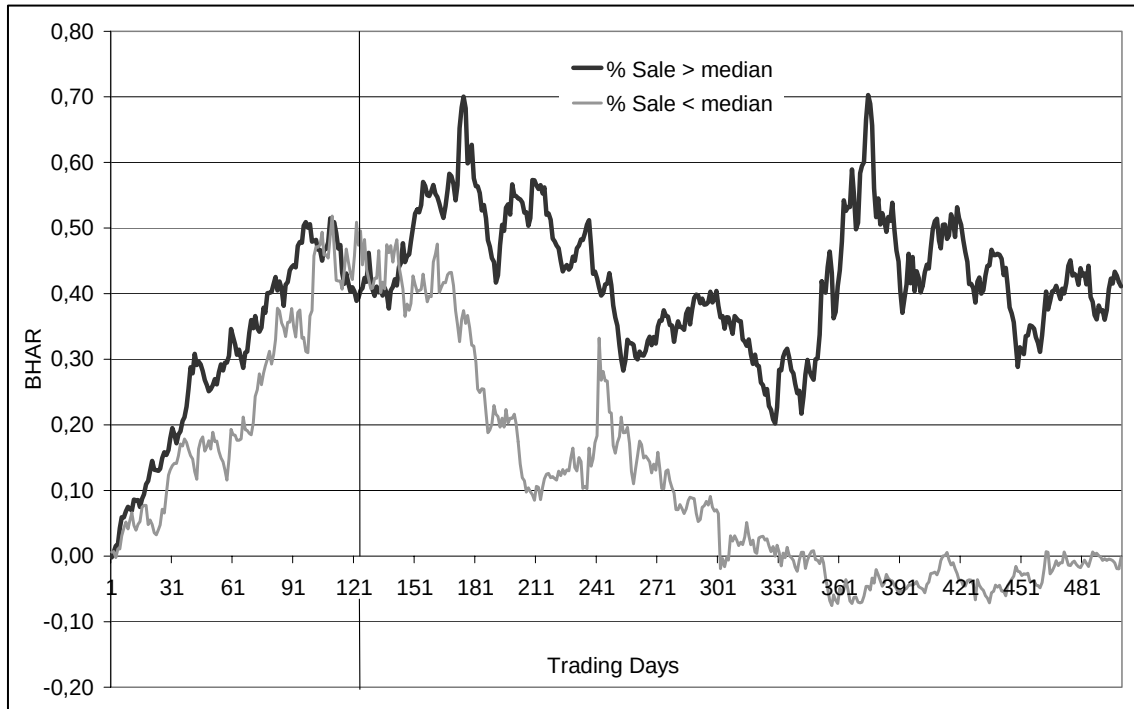


Table 7
Venture-backed IPOs and the sale of secondary shares in the primary market

This table contains the mean underpricing and buy and hold abnormal returns (BHAR) of venture-backed IPOs depending on its exit behavior at the IPO date after 123 and 500 trading days (about 6 months and 24 months, respectively). To test whether the BHAR are significantly different from zero we apply Student's skewness-adjusted t-test (t adj.) and a simple sign test (z Sign). Furthermore we test whether the mean returns of both sub-samples are significantly different applying a t-test for in in means (t means) and Wilcoxon's test of sum of ranks (U-Test). a/b/c denotes significance at the 1%/ 5% /10% level.

	N	Mean	Median	% Neg.	Std. Dev.	Skew.	Kurt.	t (adj.)	z (Sign)	t (means)	z (U-Test)
Underpricing											
Sale > Median	28	54.5%	31.0%	0.0%	0.65	1.27	3.72	6,29 ^a	5,48 ^a		
Sale < Median	29	27.4%	6.7%	24.1%	0.50	2.13	7.58	4,11 ^a	2,97 ^a	1,79 ^c	1,98 ^b
123 Trading Days											
Sale > Median	28	39.4%	16.6%	42.9%	0.82	0.80	2.59	2,88 ^a	0.94		
Sale < Median	29	47.4%	9.0%	44.8%	1.26	2.34	8.76	2,63 ^b	0.74	-0.29	0.06
500 Trading Days											
Sale > Median	28	41.1%	-2.7%	53.6%	2.20	2.29	9.08	1.14	-0.19		
Sale < Median	29	-0.3%	-6.2%	65.5%	0.65	2.61	15.15	-0.02	-1.49	0.95	0.22

Figure 8
Bank-financed IPOs and the sale of secondary shares in the primary market

The figure shows the long-run performance (BHAR) of bank-financed IPOs. The 44 bank-financed IPOs are separated by the criteria, whether the bank sells more or less than the median value of the shares in the primary market. 123 trading days correspond to 6 months and 500 trading days to 24 months of trading in the secondary market. The end of the lock-up period is shown by the vertical line.



Table 8
Bank-financed IPOs and their sale of secondary shares in the primary market

This table contains the mean underpricing and buy and hold abnormal returns (BHAR) of bank-financed IPOs depending on its exit behavior at the IPO date after 123 and 500 trading days (about 6 months and 24 months, respectively). To test whether the BHAR are significantly different from zero we apply Student's skewness-adjusted t-test (t adj.) and a simple sign test (z Sign). Furthermore we test whether the mean returns of both sub-samples are significantly different applying a t-test for differences in means (t means) and Wilcoxon's test of sum of ranks (U-Test). a/b/c denotes significance at the 1%/ 5% /10% level.

	N	Mean	Median	% Neg.	Std. Dev.	Skew.	Kurt.	t (adj.)	z (Sign)	t (means)	z (U-Test)
Underpricing											
Sale > median	22	87.8%	61.2%	0.0%	0.99	1.22	3.73	5,69 ^a	4,90 ^a	1.01	1.17
Sale < median	22	63.7%	9.1%	11.1%	0.93	1.60	5.29	4,00 ^a	4,05 ^a		
123 Trading Days											
Sale > median	22	-17.4%	-26.2%	77.3%	0.38	1.28	4.82	-2,61 ^b	-2,35 ^b	-2,84 ^a	-3,10 ^a
Sale < median	22	70.3%	22.1%	31.8%	1.40	1.96	6.25	3,14 ^a	1,92 ^a		
500 Trading Days											
Sale > median	22	-21.0%	-6.0%	68.2%	0.88	-2.79	13.47	-0.88	-1.49	-0.77	0.09
Sale < median	22	-1.2%	-7.4%	68.2%	0.82	0.44	5.52	-0.07	-1.49		

Figure 9
Bank-financed IPOs and analyst recommendations

The figure shows the long-run performance (BHAR) of bank-financed IPOs. The bank-financed IPOs are separated by the criteria of how intense the banks' analysts recommend buying the shares of the IPO in the first 12 months of trading. In 14 cases, the bank is a shareholder of the IPO and issues two or more buy recommendations. In the remaining 28 cases the bank issues none or only one buy recommendation. 123 trading days correspond to 6 months and 500 trading days to 24 months of trading in the secondary market. The end of the lock-up period is shown by the vertical line.

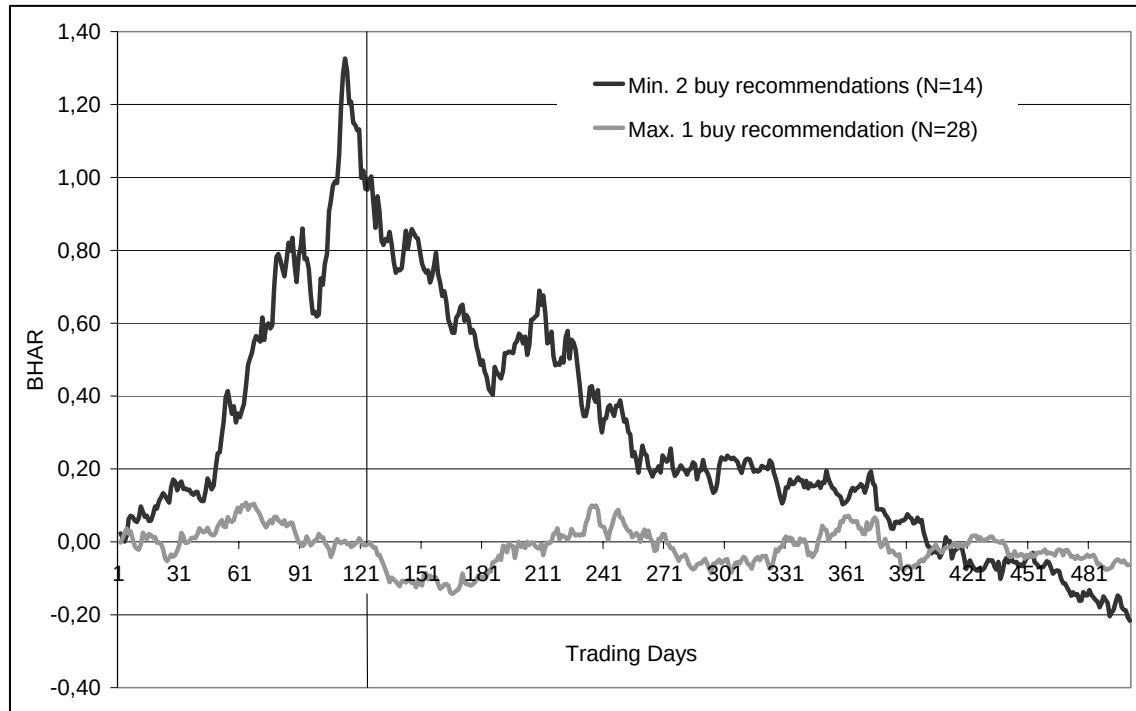


Table 9
Bank-financed IPOs and analyst recommendations

The table contains the mean underpricing and buy and hold abnormal returns (BHAR) of bank-financed IPOs depending on the number of buy recommendations published by the affiliated bank after 123 and 500 days of trading (about 6 months and 24 months, respectively). To test whether the BHAR are significantly different from zero we apply Student's skewness-adjusted t-test (t adj.) and a simple sign test (z Sign). Furthermore we test whether the mean returns of both sub-samples are significantly different applying a t-test for differences in means (t means) and Wilcoxon's test of sum of ranks (U-Test). a/b/c denotes significance at the 1%/ 5% /10% level.

	N	Mean	Median	% Neg.	Std. Dev.	Skew.	Kurt.	t (adj.)	z (Sign)	t (means)	z (U-Test)
Underpricing											
Min. 2 BR	14	51.9%	6.8%	7.1%	0.75	1.12	2.73	3,29 ^a	3,47 ^a		
Max. 1 BR	28	85.8%	61.2%	0.0%	1.00	1.45	4.31	6,45 ^a	5,48 ^a	-1.23	-1.63
123 Trading Days											
Min. 2 BR	14	86.6%	29.4%	35.7%	1.70	1.53	4.11	2,42 ^b	1.34		
Max. 1 BR	28	-0.7%	-12.1%	60.7%	0.53	1.23	4.21	-0.07	-0.94	1,88 ^c	1,73 ^c
500 Trading Days											
Min. 2 BR	14	-20.8%	-7.3%	64.3%	1.29	-1.07	5.93	-0.58	-0.80		
Max. 1 BR	28	-6.3%	-6.1%	67.9%	0.58	-0.18	7.21	-0.57	-1,70 ^c	-0.40	-0.05

Table 10
OLS cross sectional regression on IPO underpricing

The table contains the results of OLS regressions with White's heteroscedasticity-consistent standard errors. The dependent variable is IPO underpricing. All models include the control variables market value of equity (ln Firm Size) and market-to-book-ratio (ln Market/Book). The variables Venture Capital, VC Bank, VC Non-bank, VC UW Syndicate and VC Lead UW are dummy variables equal to 1 if the IPO is venture-backed, bank-financed, non-bank-financed and if the VC is underwriter or lead underwriter, respectively. VC, VC Non-bank and VC Bank ownership present their shareholdings pre IPO, Δ the change in the number of stocks at the IPO date and Lock-up the length of the lock-up period of these shareholders in months. a/b/c denotes significance at the 1%/5%/10% level.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
Intercept	-2.54 [-3.75 ^a]	-2.51 [-3.72 ^a]	-2.55 [-3.78 ^a]	-2.53 [-3.69 ^a]	-2.92 [-3.29 ^a]	-2.81 [-3.20 ^a]	-2.65 [-1.23]
ln Firm Size	0.09 [2.22 ^b]	0.09 [2.22 ^b]	0.09 [2.30 ^b]	0.09 [2.23 ^a]	0.10 [1.68 ^b]	0.10 [2.03 ^b]	0.07 [0.47]
ln Market/Book	0.83 [7.38 ^a]	0.83 [7.43 ^a]	0.82 [7.33 ^a]	0.83 [7.48 ^a]	1.01 [5.08 ^a]	0.86 [6.33 ^a]	1.20 [2.94 ^a]
Venture Capital	0.11 [1.64]						
VC Bank		0.22 [2.30 ^b]					
VC Non-bank		-0.01 [-0.19]					
VC UW Syndicate			0.27 [2.42 ^b]				
VC Lead UW				0.27 [1.87 ^c]			
VC Ownership					-0.00 [-0.77]		
Δ VC Ownership					-0.06 [-0.30]		
VC Lock-up					0.01 [0.88]		
VC Non-bank Ownership						-0.00 [-0.86]	
Δ VC Non-bank Ownership						-0.11 [-0.47]	
VC Non-bank Lock-up						0.00 [0.10]	
VC Bank Ownership							0.00 [0.56]
Δ VC Bank Ownership							0.19 [0.63]
VC Bank Lock-up							0.01 [0.69]
Adjusted R ²	0.43	0.43	0.43	0.43	0.57	0.56	0.58
# Observations	307	307	307	307	101	72	44

Table 11
OLS cross sectional regression on IPO BHAR after 123 trading days

The table contains the results of OLS regressions with White's heteroscedasticity-consistent standard errors. The dependent variable is IPO BHAR after 6 months of trading. All models include the control variables market value of equity (ln Firm Size) and market-to-book-ratio (ln Market/Book). The variables Venture Capital, VC Bank, VC Non-bank, VC UW Syndicate and VC Lead UW are dummy variables equal to 1 if the IPO is venture-backed, bank-financed, non-bank-financed and if the VC is underwriter or lead underwriter, respectively. VC, VC Non-bank and VC Bank ownership present their shareholdings pre IPO, Δ the change in the number of stocks at the IPO date and Lock-up the length of the lock-up period by these shareholders in months. a/b/c denotes significance at the 1%/ 5% /10% level.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
Intercept	0.95 [0.58]	1.13 [0.70]	1.11 [0.71]	1.30 [0.82]	0.92 [0.33]	-3.07 [-0.99]	9.93 [1.91 ^c]
ln Firm Size	-0.00 [-0.05]	-0.01 [-0.14]	-0.01 [-0.13]	-0.02 [-0.23]	0.06 [0.43]	0.21 [1.32]	-0.39 [-1.58]
ln Market/Book	-0.38 [-1.70 ^c]	-0.37 [-1.67 ^c]	-0.38 [-1.67 ^c]	-0.36 [-1.65 ^c]	-0.60 [-3.50 ^a]	-0.59 [-2.69 ^a]	-0.23 [-1.05]
Venture Capital	0.11 [0.83]						
VC Bank		0.00 [0.02]					
VC Non-bank		0.03 [0.22]					
VC UW Syndicate			0.07 [0.32]				
VC Lead UW				-0.17 [-1.03]			
VC Ownership					-0.01 [-1.06]		
Δ VC Ownership					1.02 [1.59]		
VC Lock-up					-0.04 [-1.71 ^c]		
VC Non-bank Ownership						0.00 [0.07]	
Δ VC Non-bank Ownership						-0.14 [-0.22]	
VC Non-bank Lock-up						0.04 [0.94]	
VC Bank Ownership							-0.03 [1.89 ^c]
Δ VC Bank Ownership							1.84 [2.51 ^b]
VC Bank Lock-up							-0.10 [-2.95 ^a]
Adjusted R ²	0.03	0.02	0.02	0.03	0.11	0.00	0.33
# Observations	307	307	307	307	101	72	44

Table 12
OLS cross sectional regression on IPO's BHAR after 500 trading days

The table contains the results of OLS regressions with White's heteroscedasticity-consistent standard errors. The dependent variable is IPO BHAR after 6 months of trading. All models include the control variables market value of equity (ln Firm Size) and market-to-book-ratio (ln Market/Book). The variables Venture Capital, VC Bank, VC Non-bank, VC UW Syndicate and VC Lead UW are dummy variables equal to 1 if the IPO is venture-backed, bank-financed, non-bank-financed and if the VC is underwriter or lead underwriter, respectively. VC, VC Non-bank and VC Bank ownership present their shareholdings pre IPO, Δ the change in the number of stocks at the IPO date and Lock-up the length of the lock-up period by these shareholders in months. a/b/c denotes significance at the 1%/ 5% /10% level.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
Intercept	-0.81 [-0.84]	-0.78 [-0.82]	-0.59 [-0.63]	-0.65 [-0.68]	-1.26 [-0.42]	-2.96 [-0.80]	2.16 [0.90]
ln Firm Size	0.08 [1.44]	0.08 [1.42]	0.07 [1.32]	0.07 [1.37]	0.17 [0.98]	0.27 [1.25]	-0.05 [-0.34]
ln Market/Book	-0.42 [-1.86 ^c]	-0.41 [-1.82 ^c]	-0.41 [-1.80 ^c]	-0.42 [-1.86 ^c]	-1.17 [-2.37 ^b]	-1.48 [-2.16 ^b]	-0.52 [-2.04 ^b]
Venture Capital	0.06 [0.38]						
VC Bank		-0.15 [-0.98]					
VC Non-bank		0.16 [0.90]					
VC UW Syndicate			-0.13 [-0.74]				
VC Lead UW				-0.06 [-0.39]			
VC Ownership					-0.00 [-0.21]		
Δ VC Ownership					-0.79 [-1.41]		
VC Lock-up					-0.03 [-0.72]		
VC Non-bank Ownership						-0.00 [-0.39]	
Δ VC Non-bank Ownership						-1.25 [-1.50]	
VC Non-bank Lock-up						0.02 [0.31]	
VC Bank Ownership							-0.00 [-0.34]
Δ VC Bank Ownership							0.19 [0.54]
VC Bank Lock-up							-0.05 [-1.82 ^c]
Adjusted R ²	0.02	0.02	0.02	0.02	0.12	0.14	0.07
# Observations	307	307	307	307	101	72	44